

Mass-Conserving Coagulation–Fragmentation Generative Models

Finite-Horizon Time Reversal on Variable-Size Weighted Sets

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Draft: theory and algorithms complete; experiments not yet run

Abstract

Many data objects are unordered collections with a variable number of components and a meaningful positive additive quantity, such as mass, area, energy, or probability weight. Standard diffusion models usually keep the number of components fixed, while a prominent trans-dimensional alternative removes and inserts one component at a time without an additive pair-merge constraint. We introduce a different construction based on conservative binary fragmentation and coagulation. The data-to-reference process first fragments components and moves component mass into an explicit reservoir. It then removes every remaining component, so that the terminal state is exactly an all-reservoir state. The finite-horizon reverse process starts from this simple state and performs nucleation, growth, and pairwise coagulation.

The paper makes the construction precise on an unordered, variable-dimensional state space. We define compatible reference measures, conservative split and merge maps, and a nonnegative reservoir flow. We derive the reverse jump kernel from probability flux, including the mass Jacobian and the cancellation induced by a consistent unordered-pair convention. We then specialize a path-space evidence lower bound to this conservative marked-event process. Its trainable part is the negative log likelihood of reversed forward trajectories under a leaf-count, marked-event model. This gives a simple simulation-based learning algorithm that does not require evaluating the unknown marginal density ratio appearing in the exact reverse rates. We prove terminal absorption, non-explosion, exact mass conservation, permutation invariance, Fisher consistency under stated model-class assumptions, and an endpoint error bound. We also give complete forward-simulation, training, and reverse-sampling algorithms. An optional extension replaces uniform evaporation with a spatial control field motivated by clearing a fogged mirror one region at a time. The experimental protocol is specified, but no experimental results are reported in this draft.

1 Introduction

A set-valued observation can be written informally as

$$X = \{(m_i, z_i)\}_{i=1}^N, \quad m_i > 0, \quad z_i \in \mathbb{R}^q, \quad (1)$$

where the braces are understood as an unordered multiset, the order of the elements has no meaning, and the cardinality N may change from one example to another. The scalar m_i is an additive quantity. Depending on the application, it may represent physical mass, energy, area, intensity, or a normalized weight. The vector z_i stores position and other continuous features.

We study generative models for such observations under the exact constraint

$$v + \sum_{i=1}^N m_i = M. \quad (2)$$

Here $M > 0$ is fixed or supplied as conditioning information, and $v \geq 0$ is an uncondensed reservoir. Introducing v is important. It lets a component grow or evaporate without violating the additive constraint.

The physical picture is condensation, but the method is not obtained by replacing a Gaussian noise schedule with the Hertz–Knudsen equation. The useful structure is instead the change in topology:

reservoir $\longrightarrow$ nucleation $\longrightarrow$ growth $\longrightarrow$ pairwise coagulation $\longrightarrow$ sample.

Training uses the opposite direction. A prescribed forward process fragments data components, transfers their mass to the reservoir, and finally deletes them. The learned model follows the time-reversed trajectories.

The motivating mirror observation also contains a second idea. A hair dryer does not clear the whole mirror uniformly: it acts on one small region, the user observes what remains covered, and then moves the dryer. Section 11 turns this observation into an optional spatially controlled evaporation field. The base model remains the uniform-control special case.

This paper has five main contributions.

1. We define a conservative fragmentation–evaporation process on finite unordered weighted configurations. A two-stage schedule makes the all-reservoir terminal state exact rather than approximate.
2. We specialize finite-horizon Markov-process reversal to this state space. The resulting reverse events are nucleation and pairwise coagulation. The coordinate derivation handles both the mass-dependent Jacobian and unordered-pair counting.
3. We specialize a path-space evidence lower bound to the conservative marked-event process and obtain a tractable point-process loss. Forward paths provide supervised reverse events, while a survival integral trains merge timing and stopping.
4. We give implementable neural parameterizations and event-driven algorithms with exact conservation, positivity, permutation invariance, and explicit non-explosion conditions.
5. We formulate a locally controlled evaporation extension, prove that it remains conservative and positive, and explain its conditional reverse, flow Jacobian, feedback policy, and implementation.

The contribution is deliberately narrower than “a condensation diffusion model.” Variable-dimensional generation and general jump-process time reversal already exist. Our focus is the complete conservative one-to-two/two-to-one construction on unordered weighted sets.

2 Related work

Corruption and reversal. Diffusion models learn to reverse a prescribed corruption process [16, 12, 17]. Their connection to nonequilibrium thermodynamics is therefore foundational rather than new. Cold Diffusion showed that useful models can invert degradations other than Gaussian noising [4]. Consequently, calling a degradation “evaporation” is not by itself a methodological contribution.

Trans-dimensional generation. The closest generative framework is the Jump Diffusion Model of Campbell et al. [8]. It deletes individual components in the forward direction and learns state-conditioned single-component insertion in reverse, together with a trans-dimensional reversal formula and an evidence lower bound. Learning to Jump uses thinning and thickening for latent counts [9]. The 2026 preprint Existence-Field Diffusion instead represents variable cardinality by continuous existence variables [15]. We do not claim that variable-dimensional generation is new. Our distinction is a conservative binary split whose reverse must select and merge an interacting pair.

Structured expansion and branching. Iterative Local Expansion reverses graph coarsening by repeatedly splitting coarse nodes and refining local structure [5]. The 2026 preprint Hierarchical Discrete Flow Matching also uses coarse-to-fine graph expansion [7]. In particle physics, JUNIPR generates particle sets through conservation-aware binary branchings [3]. These papers show that structured splitting is not itself new. In our construction, the genealogy is sampled by the prescribed corruption rather than fixed by a preprocessing-induced clustering tree.

Dimension-changing maps and their Jacobians also have a long history in reversible-jump Markov chain Monte Carlo [11]. Unlike that stationary MCMC setting, we study a learned finite-horizon reversal.

Coagulation, fragmentation, and reversal. Coagulation–fragmentation processes have a large mathematical literature [1, 14, 6]. General time-reversal results also exist for diffusions and Markov processes with jumps [2, 10]. We therefore do not present the probability-flux identity as a new general theorem. The

technical work is its specialization to conservative unordered configurations, including compatible measures, event coordinates, counting factors, a trainable objective, and a practical sampler.

3 State space and notation

3.1 Weighted configurations

Fix a total mass $M > 0$, a feature dimension $q \geq 1$, and a finite cardinality cap $N_{\max} \geq 1$. The feature space is $\mathcal{Z} = \mathbb{R}^q$, equipped with its Borel sets and q -dimensional Lebesgue measure λ_q . We use the shorthand

$$\lambda_q^{\otimes n}(\mathrm{d}(z_1, \dots, z_n)) \equiv \mathrm{d}z_{1:n} := \prod_{i=1}^n \lambda_q(\mathrm{d}z_i). \quad (3)$$

Thus every density in this paper that contains $\mathrm{d}z_{1:n}$ is a density for continuous features relative to product Lebesgue measure. Discrete or mixed features require a different product reference measure and are not covered by the displayed coordinate formulas. All state spaces and reference measures below are conditional on the chosen M ; this dependence is suppressed in the notation $\mathbf{X}_n$ and μ_n .

A state with n components is

$$X = (v; \{(m_i, z_i)\}_{i=1}^n), \quad v \geq 0, \quad m_i > 0, \quad z_i \in \mathbb{R}^q, \quad v + \sum_{i=1}^n m_i = M. \quad (4)$$

The braces in (4) denote an unordered multiset, not an ordinary set: identical pairs may occur with multiplicity. Formally, a state is an equivalence class of labeled tuples under permutations of the n indices. Two states are equal when one can be obtained from the other by such a permutation. We write $N(X) = n$ and

$$\mathcal{M}(X) = v + \sum_{i=1}^{N(X)} m_i. \quad (5)$$

The all-reservoir state is

$$\emptyset_M = (M; \emptyset). \quad (6)$$

It is sometimes useful to view the components as the finite measure

$$\eta_X = \sum_{i=1}^{N(X)} m_i \delta_{z_i}. \quad (7)$$

Then $\eta_X(\mathcal{Z}) = M - v$, and relabeling the components does not change η_X . The symbol η_X describes one state; it is not the reference measure used to integrate over all states. If two components have exactly the same feature z , η_X combines their masses and no longer records their multiplicity. The unordered multiset in (4) is therefore the primary representation. Under the continuous feature densities used below, exact feature collisions have measure zero.

Let $\mathbf{X}_n$ be the space of states with n components and let

$$\mathbf{X} = \bigsqcup_{n=0}^{N_{\max}} \mathbf{X}_n \quad (8)$$

be their disjoint union. For the finite-horizon process introduced in Section 5, $t \in [0, T]$ denotes forward time and $X_t \in \mathbf{X}$ is the random state at time t . Set $N_t = N(X_t)$. On the event $\{N_t = n\}$, we have $X_t \in \mathbf{X}_n$. The process N_t is piecewise constant and changes only at a split or deletion event in forward time, or at a birth or merge event in reverse time. The cap is part of the model: a forward split is disabled when $N(X) = N_{\max}$. It prevents an artificial explosion of ever smaller fragments and makes the reverse leaf budget finite. A large $N_{\max}$ has no practical effect when data and simulated paths remain far below it.

Data law, corruption law, and model law. These three laws have different roles. The raw data are sampled from an unknown law on unordered weighted sets; we do not assume that nature generated them by fragmentation. The forward fragmentation–evaporation process is a user-chosen conditional corruption law used to construct training histories. The learned reverse process is the generative model: it samples a leaf budget, birth times and marks, deterministic growth, and merge events, thereby inducing a joint endpoint law for masses and features. In particular, there is no separate uniform law for $z_{1:n}$, and no independence among components, masses, and features is assumed.

Reservoir dequantization. Suppose a nonempty raw observation has normalized positive weights

$$w = (w_1, \dots, w_n) \in \Delta_{n-1}^\circ := \left\{ w \in (0, 1)^n : \sum_{i=1}^n w_i = 1 \right\}. \quad (9)$$

Only $w_{1:n-1}$ are free, because $w_n = 1 - \sum_{i < n} w_i$. We assume the raw weights have a density in these $n - 1$ simplex coordinates and that the features have a density relative to (3). The corresponding unordered raw reference measure is

$$\nu_n^{\text{raw}}(dw \, dz) = \frac{1}{n!} \mathbf{1}\{w \in \Delta_{n-1}^\circ\} dw_{1:n-1} \, dz_{1:n}. \quad (10)$$

The displayed coordinates use any labeled representative, with w_n eliminated. Permuting the representative only changes to another simplex chart with absolute Jacobian one, and the factor $1/n!$ makes the resulting unordered density label-independent. The case $n = 1$ has no free w coordinate; its simplex is a single point. The empty outcome has no normalized weight vector and is handled separately in X_0 .

A fixed reservoir fraction ξ_0 would give $\sum_i m_i = M(1 - \xi_0)$. The mass vector would therefore lie on an $(n - 1)$ -dimensional hyperplane inside the n -dimensional mass domain. That hyperplane has zero $dm_{1:n}$ measure, so the augmented data law would be singular relative to the state reference measure defined below. A continuous random reservoir fraction supplies the missing radial dimension. Specifically, draw

$$\xi \sim q_\epsilon, \quad v = \xi M, \quad m_i = (1 - \xi)Mw_i, \quad (11)$$

where q_ϵ is a positive smooth density, relative to $d\xi$, on $I_\epsilon = [\epsilon_{\min}, \epsilon_{\max}] \subset (0, 1)$. In this paper q_ϵ is fixed and independent of (w, z) ; a data-dependent dequantization density could be used if its conditional log density were included in the bound. The resulting state has $v > 0$ and $\sum_i m_i < M$, and its law can have a full-dimensional density. The dequantization density and its Jacobian must be included when a numerical data-likelihood bound is reported. In coordinates $(w_1, \dots, w_{n-1}, \xi) \mapsto (m_1, \dots, m_n)$, with $w_n = 1 - \sum_{i < n} w_i$, the absolute Jacobian determinant (see Appendix A) is

$$M^n(1 - \xi)^{n-1}. \quad (12)$$

At the output, no auxiliary ξ needs to be stored. For a nonempty state, let $c = \sum_j m_j = M - v > 0$ and decode

$$\hat{w}_i = \frac{m_i}{c} = \frac{m_i}{\sum_j m_j}. \quad (13)$$

If we define the state-implied fraction $\hat{\xi} = v/M$, then $c = M(1 - \hat{\xi})$, so this is also $\hat{w}_i = m_i/[M(1 - \hat{\xi})]$. The empty state is a separate discrete outcome. If an application observes a genuine reservoir, no augmentation or normalization decoder is needed provided that its joint law is supported on $v > 0$ and is dominated by the interior reference measure below. An atom at $v = 0$ requires an additional boundary reference measure or an analogous dequantization.

Proposition 3.1 (The normalized encoder and decoder agree). *For every nonempty state, the numbers in (13) are positive and sum to one. If a raw normalized set is encoded by (11), decoding it recovers its original weights exactly. Conversely, decoding any nonempty state and re-encoding it with $\hat{\xi} = v/M$ recovers all of its masses and its reservoir; the features are copied unchanged in both maps.*

Proof. Because every $m_i > 0$ and $c = \sum_j m_j > 0$, each $\hat{w}_i > 0$, and $\sum_i \hat{w}_i = c/c = 1$. For an encoded datum, $c = \sum_i M(1 - \xi)w_i = M(1 - \xi)$, hence $m_i/c = w_i$. Conversely, with $\hat{\xi} = v/M$,

$$M(1 - \hat{\xi})\hat{w}_i = (M - v)\frac{m_i}{M - v} = m_i, \quad M\hat{\xi} = v.$$

Thus the two maps are inverses on every nonempty stratum. $\square$

For completeness, the raw normalized-data likelihood obeys the standard dequantization bound

$$\log p_\theta^{\text{raw}}(w_{1:n-1}, z_{1:n}) \geq \mathbb{E}_{\xi \sim q_\epsilon} [\log p_{\theta,n}(X(\xi)) + n \log M + (n-1) \log(1-\xi) - \log q_\epsilon(\xi)]. \quad (14)$$

Here p_θ^{raw} is a density relative to ν_n^{raw} , whereas p_θ is a state density relative to the measure below. More explicitly, if P_θ is the endpoint law of the learned reverse process, then $p_{\theta,n}$ is its density on $\mathbf{X}_n$. The bound follows by marginalizing the state-implied reservoir fraction over $(0, 1)$, restricting the integral to I_ϵ , inserting q_ϵ/q_ϵ , and applying Jensen's inequality. The path ELBO below can be substituted for $\log p_{\theta,n}(X(\xi))$ to obtain one combined lower bound. Appendix A gives the complete derivations of (12) and (14). Throughout the theory below, P_{data} denotes the law of this augmented state X (or the directly observed state when a reservoir is available).

3.2 Reference measure

An unordered density requires a counting convention. Let $\mathbf{q}_n$ map a labeled coordinate tuple $(m_{1:n}, z_{1:n})$ to its unordered state, with $v = M - \sum_i m_i$. We define μ_n by requiring that, for every nonnegative measurable function f , where $\text{d}m_{1:n} := \prod_{i=1}^n \text{d}m_i$ is n -dimensional Lebesgue volume,

$$\int_{\mathbf{X}_n} f(X) \mu_n(\text{d}X) = \frac{1}{n!} \int f(\mathbf{q}_n(m_{1:n}, z_{1:n})) \mathbf{1}\left\{m_i > 0, \sum_{i=1}^n m_i < M\right\} \text{d}m_{1:n} \text{d}z_{1:n}. \quad (15)$$

We use the corresponding shorthand $\mu_n(\text{d}X) = n!^{-1} \mathbf{1}\{m_i > 0, \sum_i m_i < M\} \text{d}m_{1:n} \text{d}z_{1:n}$.

For fixed M , the mass-coordinate domain is the open scaled simplex

$$D_n(M) = \left\{m \in (0, \infty)^n : \sum_{i=1}^n m_i < M\right\}. \quad (16)$$

Each individual mass lies in $(0, M)$, but the joint domain is not the full box $(0, M)^n$ because the masses share the same budget. Lebesgue volume on $D_n(M)$ is a reference measure, not a uniform probability distribution.

The condition $m_i > 0$ repeats the definition of $\mathbf{X}_n$ in ambient Euclidean coordinates so that the domain of integration is explicit. A zero-mass element is absent and belongs to a lower-cardinality stratum. The strict inequality selects the interior $v > 0$. We keep the boundary $v = 0$ in the state space for event-map and flow statements, but it has zero μ_n measure; a law with an atom on that boundary would need an additional singular measure component.

The factor $1/n!$ is needed because a generic unordered configuration has $n!$ labeled coordinate representations. Integrating a symmetric function over all labeled coordinates without this factor would count the same state $n!$ times. The state-specific finite measure η_X in (7) does not solve this counting problem: it represents one configuration, whereas μ_n defines volume across all configurations. For example, when $n = 2$, the labeled tuples $((m_1, z_1), (m_2, z_2))$ and $((m_2, z_2), (m_1, z_1))$ describe the same unordered state. The factor $1/2!$ counts that state once. Equivalently, one could integrate only over one canonically ordered chamber and omit the factorial. Feature collisions can reduce the number of distinct labelings, but those collisions form a null set under product Lebesgue measure. We write

$$\mu = \sum_{n=0}^{N_{\max}} \mu_n, \quad (17)$$

with μ_0 a unit point mass at $\emptyset_M$. For later use, write $P_t = \mathcal{L}(X_t)$ for the law of X_t . Whenever the restriction of P_t to $\mathbf{X}_n$ is absolutely continuous with respect to μ_n , define the stratum density by

$$p_{t,n} = \frac{\text{d}(P_t|_{\mathbf{X}_n})}{\text{d}\mu_n}, \quad P_t(\text{d}X) = p_{t,n}(X) \mu_n(\text{d}X), \quad X \in \mathbf{X}_n. \quad (18)$$

It integrates to the probability of the corresponding cardinality,

$$\int_{\mathbf{X}_n} p_{t,n}(X) \mu_n(\text{d}X) = P_t(\mathbf{X}_n) = \mathbb{P}(N_t = n), \quad (19)$$

and therefore need not integrate to one.

The element $dz_{1:n}$ in (15) is a volume element, not a probability distribution. A probability law is obtained only after multiplying μ_n by the joint density $p_{t,n}$; that density may couple all masses and features. During reservoir dequantization the observed $z_{1:n}$ are copied unchanged, so their Jacobian block is the identity. We integrate out the auxiliary ξ in the raw likelihood, but we do not integrate out $z_{1:n}$ because the features are observed. A weights-only marginal would additionally integrate over $\mathcal{Z}^n$ with respect to $\lambda_q^{\otimes n}$.

For any measurable $A \subseteq \mathbf{X}$, the complete law is reconstructed from its stratum densities as

$$P_t(A) = \sum_{n=0}^{N_{\max}} \int_{A \cap \mathbf{X}_n} p_{t,n}(X) \mu_n(dX). \quad (20)$$

Thus μ_n is a sigma-finite coordinate-volume measure, not a uniform probability law on configurations.

The $1/n!$ convention is not cosmetic. It determines the factors in the birth and merge rates derived in Section 6.

3.3 A notation table

Symbol	Meaning
M	conserved total mass, fixed or conditioned upon
$\mathcal{Z}, \lambda_q$	continuous feature space $\mathbb{R}^q$ and its Lebesgue reference measure
$N_{\max}$	maximum allowed number of components
v	mass in the reservoir
(m_i, z_i)	mass and continuous feature of component i
$N(X), N_t$	number of components in state X , and $N(X_t)$ at time t
η_X	finite measure representing one weighted configuration
$\mathbf{X}_n, \mathbf{X}$	n -component state space and its disjoint union
μ_n	unordered reference measure with factor $1/n!$
ν_n^{raw}	reference measure for unordered normalized raw data
Δ_{n-1}°	open simplex of n positive normalized weights
q_e	fixed density of the auxiliary reservoir fraction ξ
$P_t, p_{t,n}$	forward marginal law and its density on $\mathbf{X}_n$
$P_\theta, p_{\theta,n}$	learned endpoint law and its density on $\mathbf{X}_n$
y	optional conditioning information
T, τ, S_N	terminal time, switch time, and nucleation duration $S_N = T - \tau$
β, γ, δ	prescribed split, reservoir-flow, and deletion schedules
κ_t	prescribed normalized split-mark density
$\mathcal{S}_i, \mathcal{C}_{ab}$	split of component i and merge of pair a, b
$\mathcal{D}_i, \mathcal{B}_{m,z}$	deletion and insertion maps
$Q_t, \bar{Q}_s$	forward and reverse jump kernels
$\nu_X, \underline{q}$	reference measures for forward histories and raw reverse records
$Y_s, \tilde{X}_s$	exact reversed state and reversed training path at reverse time s
L, Π_θ	Stage-N leaf budget and its learned probability mass function
$A_\theta^{(\omega)}, A_\theta^{(m)}$	birth-mark densities in mass-fraction and absolute-mass coordinates
B_L	fixed conditional birth hazard determined by L
Λ_θ^C	learned total coagulation rate
$K_\theta(a, b \mid X, s)$	learned merge intensity for unordered pair $a < b$
$a_t(x)$	optional control field at spatial coordinate $x = \text{pos}(z)$
$\lambda_i(t, X, a_t)$	controlled evaporation rate of component i

4 Conservative event maps

This section defines the four operations used by the model. Each operation preserves $\mathcal{M}(X) = M$ exactly.

4.1 Mass exchange with the reservoir

Between jumps, component features stay fixed and masses obey

$$\frac{dm_i}{dt} = -\gamma(t) \frac{v}{M} m_i, \quad (21)$$

$$\frac{dv}{dt} = \gamma(t) \frac{v}{M} \sum_{i=1}^{N(X)} m_i, \quad (22)$$

where $\gamma(t) \geq 0$ is prescribed. This is the forward evaporation flow. It is deliberately simple: it transfers mass to the reservoir without changing relative component masses.

Let $c = M - v = \sum_i m_i$ and $r = c/(M - c) = c/v$. On an interval with no jump,

$$r(t) = r(t_0) \exp\left(-\int_{t_0}^t \gamma(u) du\right), \quad c(t) = M \frac{r(t)}{1 + r(t)}, \quad (23)$$

and every m_i is multiplied by $c(t)/c(t_0)$. Thus the flow is available in closed form. To verify the formula, sum (21) to obtain

$$\frac{dc}{dt} = -\gamma(t) \frac{c(M - c)}{M}. \quad (24)$$

For $0 < c < M$, differentiating $\log r = \log c - \log(M - c)$ gives $d \log r / dt = -\gamma(t)$, which integrates to (23). The two boundary cases are fixed points: $c = 0$ is the empty state and $v = M$, while $v = 0$ means $c = M$. The odds r need not be evaluated at either boundary.

Running time backward changes the signs:

$$\frac{dm_i}{ds} = +\gamma(T - s) \frac{v}{M} m_i, \quad \frac{dv}{ds} = -\gamma(T - s) \frac{v}{M} \sum_i m_i. \quad (25)$$

Equation (25) is the growth stage of generation. Because its vector field vanishes at $m_i = 0$ and $v = 0$, neither component mass nor reservoir mass can cross below zero.

4.2 Binary fragmentation and coagulation

Choose a component (m, z) , a fraction $\rho \in (0, 1)$, and a displacement $u \in \mathbb{R}^q \setminus \{0\}$. Define

$$m_a = \rho m, \quad z_a = z - (1 - \rho)u, \quad (26)$$

$$m_b = (1 - \rho)m, \quad z_b = z + \rho u. \quad (27)$$

The split map $\mathcal{S}_i(X; \rho, u)$ replaces component i by these two children. It preserves both mass and the mass-weighted feature centroid:

$$m_a + m_b = m, \quad m_a z_a + m_b z_b = m z. \quad (28)$$

The inverse map merges an unordered pair $a < b$:

$$\mathcal{C}_{ab}(X) = X \setminus \{(m_a, z_a), (m_b, z_b)\} \cup \left\{ \left(m_a + m_b, \frac{m_a z_a + m_b z_b}{m_a + m_b} \right) \right\}. \quad (29)$$

To avoid counting the same split twice, let $\mathcal{U}_+$ contain exactly one of u and $-u$ for almost every nonzero u . One choice is the set in which the first nonzero coordinate is positive. We take split marks in

$$\mathcal{A} = (0, 1) \times \mathcal{U}_+. \quad (30)$$

Given two distinct children, canonical orientation determines $u = z_b - z_a \in \mathcal{U}_+$ and then

$$m = m_a + m_b, \quad \rho = \frac{m_a}{m}, \quad z = \frac{m_a z_a + m_b z_b}{m}. \quad (31)$$

4.3 Deletion and nucleation

The deletion map transfers an entire component to the reservoir:

$$\mathcal{D}_i(X) = (v + m_i; \{(m_j, z_j) : j \neq i\}). \quad (32)$$

Its inverse is a birth or nucleation event. For $0 < m \leq v$,

$$\mathcal{B}_{m,z}(X) = \left(v - m; \{(m_i, z_i)\}_{i=1}^{N(X)} \cup \{(m, z)\} \right). \quad (33)$$

The equality case $m = v$ maps to the boundary $v = 0$. All coordinate-density formulas below use the interior case $0 < m < v$. With reservoir-dequantized data, finite flow times, and split fractions in $(0, 1)$, the forward law stays in that interior before deletion; reversing a deletion then indeed has $m < v$ because the pre-deletion reservoir was strictly positive. Unlike a model restricted to single-component insertion without a pair-merge operation, births here are followed by learned pairwise merges. The representation of a final component can therefore be assembled from several nucleated pieces.

Proposition 4.1 (Exact conservation of every operation). *The flows in (21)–(22) and (25), and the maps $\mathcal{S}_i$, $\mathcal{C}_{ab}$, $\mathcal{D}_i$, and $\mathcal{B}_{m,z}$ all preserve $\mathcal{M}(X) = M$.*

Proof. For either flow, summing the derivatives of v and all m_i gives zero. A split replaces m by $\rho m + (1 - \rho)m = m$, and a merge performs the reverse operation. A deletion adds m_i to v before removing m_i ; a birth subtracts the same amount from v before inserting it. Therefore every operation leaves the total unchanged. $\square$

4.4 A complete numerical example

Take $M = 10$ and $q = 1$, so each z_i is a scalar position. Consider the initial state

$$X_0 = (2; \{(3, 1), (5, 4)\}) \in \mathbf{X}_2. \quad (34)$$

Here the reservoir has mass 2, the two components have masses 3 and 5, and their positions are 1 and 4. The total is $2 + 3 + 5 = 10 = M$.

Split the component $(m, z) = (5, 4)$ using $\rho = 0.4$ and $u = 2$. Equations (26)–(27) give

$$(5, 4) \mapsto (2, 2.8), (3, 4.8). \quad (35)$$

The resulting state is

$$X_{t_1} = (2; \{(3, 1), (2, 2.8), (3, 4.8)\}) \in \mathbf{X}_3. \quad (36)$$

The split preserves mass, and it also preserves the weighted feature because $2(2.8) + 3(4.8) = 5(4) = 20$.

Suppose a subsequent flow segment reduces the total component mass from 8 to 6. The basic flow preserves relative masses, so it multiplies each component mass by $6/8 = 3/4$ and transfers the remaining mass to the reservoir:

$$X_{t_2} = (4; \{(2.25, 1), (1.5, 2.8), (2.25, 4.8)\}) \in \mathbf{X}_3. \quad (37)$$

The continuous flow changes v and the masses but not the cardinality or features. If the component $(1.5, 2.8)$ is then deleted, its mass enters the reservoir and

$$X_{t_3} = (5.5; \{(2.25, 1), (2.25, 4.8)\}) \in \mathbf{X}_2. \quad (38)$$

Every displayed state has total mass 10. Reversing these operations first births $(1.5, 2.8)$, then reverses the flow, and finally merges $(2, 2.8)$ with $(3, 4.8)$. The merged component has

$$m = 2 + 3 = 5, \quad z = \frac{2(2.8) + 3(4.8)}{5} = 4, \quad (39)$$

so the reverse sequence recovers (34) exactly.

The continuous flow and four jump operations move between strata as follows:

Operation	Stratum change	Quantities changed directly
Continuous flow	$X_n \rightarrow X_n$	v, m_i
Split	$X_n \rightarrow X_{n+1}$	one component becomes two
Deletion	$X_n \rightarrow X_{n-1}$	one component moves into v
Birth	$X_n \rightarrow X_{n+1}$	mass moves out of v
Merge	$X_n \rightarrow X_{n-1}$	two components become one

5 The forward fragmentation–evaporation process

5.1 Two stages

Choose $0 < \tau < T$. The forward process starts at $X_0 \sim P_{\text{data}}$ and has two stages.

Stage F: fragmentation, $0 \leq t \leq \tau$. Each component splits at rate $\beta(t)$, where $0 \leq \beta(t) \leq \beta_{\max} < \infty$, provided that $N(X) < N_{\max}$. Conditional on a split, the parent is therefore uniform among the current components. The split mark has a known normalized density

$$\kappa_t(\rho, u \mid X, i) \quad \text{on } \mathcal{A}. \quad (40)$$

A “density on $\mathcal{A}$ ” means a nonnegative function integrating to one relative to $d\rho \lambda_q(du)$ restricted to $\mathcal{U}_+$. It may depend jointly on the full state and the selected parent; no factorization of ρ and u is assumed. A simple choice is a symmetric Beta density for ρ and a half-space Gaussian density for u . The evaporation flow (21)–(22) acts between events. There are no deletion events in this stage.

Stage E: evaporation and deletion, $\tau < t < T$. Splitting is turned off. Every remaining component is deleted with hazard

$$\delta(t) = \frac{1}{T - t}. \quad (41)$$

The conservative evaporation flow continues between deletions. Because

$$\exp\left(-\int_{\tau}^t \delta(u) du\right) = \frac{T - t}{T - \tau}, \quad (42)$$

the deletion time of each component alive at τ is uniform on (τ, T) . This observation gives an efficient exact simulator.

5.2 Jump generator

For a bounded test function $f : X \rightarrow \mathbb{R}$, the jump part of the forward generator is

$$\begin{aligned} (\mathcal{J}_t f)(X) = & \mathbf{1}\{t \leq \tau, N(X) < N_{\max}\} \sum_{i=1}^{N(X)} \beta(t) \int_{\mathcal{A}} [f(\mathcal{S}_i(X; \rho, u)) - f(X)] \kappa_t(\rho, u \mid X, i) d\rho du \\ & + \mathbf{1}\{t > \tau\} \sum_{i=1}^{N(X)} \delta(t) [f(\mathcal{D}_i(X)) - f(X)]. \end{aligned} \quad (43)$$

The full generator adds the directional derivative along the deterministic mass flow. This makes X_t a time-inhomogeneous piecewise-deterministic Markov process.

Theorem 5.1 (Well-posedness and exact terminal state). *Suppose $N(X_0) \leq N_{\max}$ almost surely, $\beta(t) \leq \beta_{\max} < \infty$, and $\int_0^T \gamma(t) dt < \infty$. Then:*

1. *the forward process has finitely many jumps on every interval $[0, t]$ with $t < T$;*
2. *masses and reservoir mass remain nonnegative and sum to M ;*

3. $X_t \rightarrow \emptyset_M$ almost surely as $t \uparrow T$. We may therefore set $X_T = \emptyset_M$.

Proof. During Stage F, a state with n components has total split rate at most $n\beta_{\max} \leq N_{\max}\beta_{\max}$. Moreover, each split raises cardinality by one and splitting stops at $N_{\max}$. Thus there are at most $N_{\max} - N(X_0)$ splits.

At the boundary $m_i = 0$, the derivative in (21) is zero. At $v = 0$, both derivatives in (21)–(22) are zero. Hence the flow cannot leave the nonnegative simplex. Proposition 4.1 shows that jumps also preserve the simplex and total mass.

There are finitely many components at time τ . By (42), each one has probability zero of surviving until T . A finite union of probability-zero survival events still has probability zero. Consequently all components are deleted before T almost surely, and every deletion transfers its remaining mass to v . The terminal state is therefore $(M; \emptyset)$. $\square$

Remark 5.2 (Why the singular-looking hazard is useful). The hazard $1/(T - t)$ diverges, but every realized deletion time is strictly smaller than T almost surely. The process has only finitely many components in Stage E, so it still has finitely many deletion events. The divergence solves a real modeling problem: a bounded deletion hazard would leave a nonzero probability of a nonempty terminal state.

6 Finite-horizon time reversal

6.1 The measure-level formula

The safest reversal formula is written using measures, before choosing coordinates.

Definition 6.1 (Forward jump flux). Let $Q_t(X, dY)$ be the forward jump-rate kernel. The jump flux at time t is the measure

$$\Gamma_t(dX, dY) = P_t(dX)Q_t(X, dY). \quad (44)$$

It describes probability per unit time moving from X to Y .

Assumption 6.2 (Regularity for finite-horizon reversal). *The state space is standard Borel. The forward process is a Borel-right, non-explosive piecewise-deterministic Markov process. Within each stratum, the flow before it reaches a boundary is a measurable C^1 bijection. Interior jumps are not simultaneous and do not occur at predictable times. The transposed incoming jump flux is absolutely continuous with respect to the current marginal law, and the marginal laws and fluxes are regular in time wherever the rates below are evaluated.*

Theorem 6.3 (Finite-horizon reverse jump kernel). *Under Assumption 6.2, define $\overline{Q}_{T-t}$ by the disintegration*

$$P_t(dY)\overline{Q}_{T-t}(Y, dX) = P_t(dX)Q_t(X, dY). \quad (45)$$

Set $Y_0 = X_T$, $Y_T = X_0$, and, for $0 < s < T$, use the càdlàg reversal

$$Y_s = X_{(T-s)-}. \quad (46)$$

Then Y is Markov. Between jumps it follows the inverse deterministic flow, and its jump-rate kernel is $\overline{Q}_s$.

Proof sketch. Fix t and a small $h > 0$. To first order in h , the joint measure of a forward jump from X at time $t - h$ to Y at time t is

$$h P_{t-h}(dX)Q_{t-h}(X, dY) + o(h).$$

Under the stated time regularity, replacing both P_{t-h} and Q_{t-h} by their time- t versions changes only the $o(h)$ term. Bayes' rule conditions this joint measure on the endpoint Y . The resulting conditional rate is exactly the disintegration in (45). When no jump occurs, a forward deterministic segment is traversed backward, so its velocity changes sign. Applying the same argument at each time proves the Markov property and identifies the reverse characteristics. The stated regularity justifies the disintegration and passage from the short-time identity to the Markov characteristics; see [10] for a general treatment. The left limit in (46) makes the reversed path right-continuous: at a forward jump time t , the reverse pre-jump state is X_t and the reverse post-jump state is X_{t-} . $\square$

Equation (45) is a finite-time statement. It depends on the current marginal P_t . It is not the same as equilibrium detailed balance.

Remark 6.4 (The terminal point mass is an entrance law). At $t = T$, the forward law is the point mass at $\emptyset_M$, so stratum-wise density ratios are used only for $0 < t < T$. The exact reverse at $s = 0$ is understood as the entrance-law limit of $X_{(T-s)-}$ as $s \downarrow 0$. This limit is unambiguous even though the coordinate density formula is not defined at the point mass itself. A numerical implementation may instead start at reverse time $s = \varepsilon$ from $\emptyset_M$, where $0 < \varepsilon < T - \tau$. By (42) and a union bound,

$$\text{TV}(P_{T-\varepsilon}, \delta_{\emptyset_M}) = \mathbb{P}(N_{T-\varepsilon} > 0) \leq \frac{\mathbb{E}[N(X_\tau)] \varepsilon}{T - \tau} \leq \frac{N_{\max} \varepsilon}{T - \tau}. \quad (47)$$

Markov kernels contract total variation, so the same quantity bounds the endpoint error caused solely by this ε -start approximation when the remaining reverse dynamics are exact.

6.2 Coordinate Jacobian for a split

For one parent, the coordinate map is

$$(m, \rho, z, u) \mapsto (m_a, m_b, z_a, z_b) \quad (48)$$

with definitions (26)–(27).

Lemma 6.5 (Split Jacobian). *The absolute determinant of the above map is*

$$\left| \frac{\partial(m_a, m_b, z_a, z_b)}{\partial(m, \rho, z, u)} \right| = m. \quad (49)$$

Proof. The mass block $(m, \rho) \mapsto (\rho m, (1 - \rho)m)$ has absolute determinant m . For each feature coordinate, the map $(z, u) \mapsto (z - (1 - \rho)u, z + \rho u)$ has matrix

$$\begin{pmatrix} 1 & -(1 - \rho) \\ 1 & \rho \end{pmatrix}$$

and determinant one. The full derivative is block lower triangular because the child masses do not depend on (z, u) . Multiplying the determinants gives m . $\square$

Assumption 6.6 (Densities for coordinate formulas). *For every relevant n and $0 < t < T$, the marginal restricted to a stratum is dominated by its reference measure:*

$$P_t|_{\mathbf{X}_n} \ll \mu_n. \quad (50)$$

Its density $p_{t,n}$ is positive at every state where a density ratio is evaluated. The measure identity in Theorem 6.3 does not need this assumption; only the coordinate ratios below do. The random reservoir augmentation in (11) is one way to avoid a singular data law.

The assumption rules out atoms on $v = 0$ and singular feature laws. It is preserved by the intended interior construction under ordinary smoothness conditions: for a finite flow interval, $m_i(t) = m_i(t_0) \exp[-\int \gamma(u)v(u)/M \, du] > 0$; an initially positive reservoir remains positive; split fractions lie in $(0, 1)$; and continuous split and birth marks hit $u = 0$ or exact feature collisions only on null sets. Absolute continuity nevertheless remains an explicit assumption, because arbitrary data or mark kernels need not satisfy it.

6.3 Explicit birth and merge rates

Let $s = T - t$ denote reverse time. We now write the exact reverse rates under the reference convention (15).

Proposition 6.7 (Exact reverse of deletion). *Let $Y \in \mathbf{X}_{n-1}$ and let $X = \mathcal{B}_{m,z}(Y) \in \mathbf{X}_n$. Define $d_t(X; m, z)$ as the forward per-component deletion intensity of the component (m, z) in X ; in the present model it is $\mathbf{1}\{t > \tau\}\delta(t)$. Then the reverse birth intensity, as a density with respect to $dm \lambda_q(dz)$, is*

$$a_s^*(m, z \mid Y) = d_t(X; m, z) \frac{p_{t,n}(X)}{p_{t,n-1}(Y)}, \quad t = T - s. \quad (51)$$

There is no extra factor of n : the sum over the n possible deleted labels cancels the ratio between $1/n!$ and $1/(n-1)!$.

Proof. Consider all labeled representations of X and sum the forward deletion flux over the deleted index. Symmetry makes the n terms equal, while the reference measure contributes $1/n!$. Thus $n/n! = 1/(n-1)!$, which is precisely the factor in the reference measure of Y . The coordinate transformation $(Y, m, z) \mapsto X$ has determinant one. Substituting these facts into (45) gives (51). $\square$

Proposition 6.8 (Exact reverse of fragmentation). *Let $X \in \mathbf{X}_n$ with $n < N_{\max}$ and let $Y = \mathcal{S}_i(X; \rho, u) \in \mathbf{X}_{n+1}$. Suppose the forward marked split intensity is*

$$r_t^{\text{split}}(X, i, \rho, u) = \beta(t) \kappa_t(\rho, u \mid X, i) \quad (52)$$

with canonical $u \in \mathcal{U}_+$. Let $a < b$ be the resulting pair in Y and let $m = m_a + m_b$. Then the exact reverse merge intensity for this unordered pair is

$$k_s^*(a, b \mid Y) = \frac{1}{m} r_t^{\text{split}}(X, i, \rho, u) \frac{p_{t,n}(X)}{p_{t,n+1}(Y)}, \quad t = T - s. \quad (53)$$

No additional cardinality or factor-of-two correction appears under the canonical-orientation convention. The formula is stated on the full-measure set $z_a \neq z_b$; the excluded diagonal corresponds to $u = 0$.

Proof. After selecting the parent, the forward reference factor is $n/n! = 1/(n-1)!$. If child coordinates are first integrated as an ordered pair, selecting an unordered pair contributes $\binom{n+1}{2}/(n+1)! = 1/[2(n-1)!]$. However, an unordered pair has two ordered representations in the labeled-coordinate integral. That full integral is twice its representation on the canonical half-space $u \in \mathcal{U}_+$, which cancels the factor $1/2$. Thus both the selected-parent measure and the canonically oriented selected-pair measure carry $1/(n-1)!$. By Lemma 6.5, $dm_a dm_b dz_a dz_b = m dm d\rho dz du$. Equating the two flux densities in (45) therefore multiplies the forward marked intensity by $1/m$. The remaining factor is the marginal density ratio $p_{t,n}(X)/p_{t,n+1}(Y)$. $\square$

The density ratios in (51) and (53) are not available for real data. Section 7 derives a simulation-based objective whose population optimum recovers the same reverse event law under the joint realizability and support assumptions of Theorem 7.3, without evaluating these ratios.

6.4 Why detailed balance is not the derivation

For a time-homogeneous process with stationary law π , detailed balance is the equality of measures

$$\pi(dX)Q(X, dY) = \pi(dY)Q(Y, dX). \quad (54)$$

For example, applying the same split coordinates as above would give

$$k^{\text{merge}}(Y, a, b) = \frac{1}{m_a + m_b} r^{\text{split}}(X, \rho, u) \frac{\pi_n(X)}{\pi_{n+1}(Y)}. \quad (55)$$

After expressing mass and feature coordinates in fixed reference units, if $\pi_n(X) \propto \exp[-\beta_{\text{th}} F_n(X)]$ relative to the same stratum measures, then

$$\log \frac{k^{\text{merge}}(Y, a, b)}{r^{\text{split}}(X, \rho, u)} = \beta_{\text{th}} \{F_{n+1}(Y) - F_n(X)\} - \log \left(\frac{m_a + m_b}{m_0} \right), \quad (56)$$

where $m_0 > 0$ is a reference mass; its constant contribution can be absorbed into the rate convention. This expression is still subject to any explicitly chosen base-measure terms. Our process is instead time-inhomogeneous and transports a data distribution to an absorbing state. Its reverse depends on $p_{t,n}$, not on a stationary Gibbs law. A learned free energy may still regularize rates, but it cannot replace the finite-horizon flux calculation.

7 Learning from reversed forward paths

7.1 Latent histories

A forward history H contains

- all split times, selected parents, and split marks (ρ, u) ;
- all deletion times and the deleted component states just before deletion;
- the deterministic flow segments between events;
- persistent lineage identifiers, used only to identify inverse merge pairs.

Conditional on $(X, \text{event times, split marks, and discrete choices})$, the recorded pre-deletion component states and flow segments are deterministic. They are stored for implementation convenience but are not additional coordinates of the history reference measure. By Theorem 5.1, every history ends at $\emptyset_M$. Reverse the event order and replace every deletion by a birth and every split by a merge. The resulting history is denoted $R(H)$. It begins at $\emptyset_M$ and ends at the original data state.

Let Q_X^F be the forward-history law conditional on $X_0 = X$, with density $q_X^F(H)$ relative to a history reference measure $\nu_X(dH)$. It is important to reverse the *joint* coordinates (X, H) , not H alone. More precisely, the history space is a disjoint union over the finite numbers of events and their discrete lineage patterns. On each piece, ν_X is Lebesgue measure on the ordered event-time simplices and continuous split marks, multiplied by counting measure on parent, deletion, and lineage choices. The raw reverse-record space is defined in the same way: its reference measure $\varrho(dR)$ is Lebesgue measure on ordered birth and merge times and continuous birth marks, multiplied by counting measure on the event counts and merge-pair choices.

Choose a fixed Borel ordering of component coordinates, such as lexicographic order, away from its measure-zero set of ties. On this canonical chamber, the quotient measure μ_n is ordinary coordinate volume: the $n!$ labeled chambers in (15) cancel its factor $1/n!$. New lineages are assigned by event order and canonical child orientation. Arbitrary lineage renamings are therefore not additional outcomes. The factor $1/n!$ occurs only in μ_n . Likewise, ϱ is plain volume on the ordered birth-time simplex; the factor $L!$ below is part of its probability density and occurs only once. Define

$$\Psi : (X, H) \mapsto R, \quad (57)$$

where R is the raw reverse event record, including event times, birth marks, and merge choices. The endpoint coordinates in X become the birth marks missing from a conditional forward history.

This resolves a simple dimension check. Suppose X has n components, the forward history has C splits, and therefore $L = n + C$ deletions. With discrete lineages fixed, the continuous dimension on the forward side is

$$n(q + 1) + C(q + 2) + L, \quad (58)$$

while the reverse record has dimension

$$L(q + 2) + C. \quad (59)$$

These are equal because $L = n + C$. On each fixed lineage pattern, Ψ is a square, almost-everywhere differentiable bijection. Its absolute Jacobian is denoted $J_\Psi(X, H)$. Appendix C gives an explicit product formula.

Lemma 7.1 (Joint history change of variables). *Outside simultaneous events, feature collisions, and coordinate-ordering ties, the map Ψ is a bijection from admissible endpoint-forward-history pairs to admissible raw reverse records. If $\mathcal{L}_\theta$ is a normalized density relative to ϱ , then*

$$\sum_{n=0}^{N_{\max}} \int_{X_n} \int \mathcal{L}_\theta(\Psi(X, H)) J_\Psi(X, H) \nu_X(dH) \mu_n(dX) = 1. \quad (60)$$

Proof. Given (X, H) , reverse the event order, turn each deletion into a birth, and turn each split into a merge. This constructs one record R . Conversely, simulate any admissible R from $\emptyset_M$ to obtain its endpoint X ;

then reverse every deterministic segment and event to recover H . The canonical component ordering and lineage rules make these constructions inverse to each other. The excluded cases are null under the continuous reference measures.

On a fixed lineage piece, ordinary multivariate change of variables gives the integrand in (60); the dimensions match by the calculation above. Summing first over its discrete patterns and then over n gives $\int \mathcal{L}_\theta(R) \varrho(\mathrm{d}R) = 1$. Because μ_n equals coordinate volume on the canonical chamber, no extra $n!$ appears. Because event times are already ordered in ϱ , no second $L!$ appears. $\square$

7.2 Path-space evidence lower bound

Theorem 7.2 (Path-space ELBO). *Let $\mathcal{L}_\theta(R)$ be the density of a raw reverse event record relative to ϱ . For $X \in \mathbf{X}_n$, define the joint endpoint-history density, relative to $\mu_n(\mathrm{d}X)\nu_X(\mathrm{d}H)$, by*

$$p_{\theta,n}(X, H) = \mathcal{L}_\theta(\Psi(X, H)) J_\Psi(X, H). \quad (61)$$

Let $p_{\theta,n}(X) = \int p_{\theta,n}(X, H) \nu_X(\mathrm{d}H)$, including the sum over discrete lineage patterns. Lemma 7.1 shows that these are the stratum densities of a normalized endpoint law. Then, whenever the supports are compatible,

$$\log p_{\theta,n}(X) \geq \mathbb{E}_{H \sim Q_X^F} [\log \mathcal{L}_\theta(\Psi(X, H)) + \log J_\Psi(X, H) - \log q_X^F(H)]. \quad (62)$$

Equality holds when Q_X^F is the model posterior over H given X .

Proof. Since q_X^F is normalized,

$$\begin{aligned} p_{\theta,n}(X) &= \int p_{\theta,n}(X, H) \nu_X(\mathrm{d}H) \\ &= \mathbb{E}_{H \sim Q_X^F} \left[\frac{p_{\theta,n}(X, H)}{q_X^F(H)} \right]. \end{aligned}$$

Insert (61), take logarithms, and apply Jensen's inequality. Equality requires the importance ratio to be constant, which is exactly the posterior condition. $\square$

In the core model with prescribed flow and corruption, only $\log \mathcal{L}_\theta$ depends on θ . Thus training needs only sampled forward histories and the trainable reverse-record likelihood. A numerical ELBO additionally uses q_X^F , J_Ψ , and, for normalized raw data, the dequantization terms from (11).

7.3 Learned reverse event model

Reverse time has two stages in the opposite order.

Auxiliary leaf budget. Let

$$L = N(X_\tau) \in \{0, \dots, N_{\max}\} \quad (63)$$

be the number of components present when forward deletion begins. The generative model first samples

$$L \sim \Pi_\theta(L \mid M, y), \quad (64)$$

where y denotes optional conditioning information. This is an auxiliary pre-coagulation count, not the final output cardinality.

Stage N: nucleation, $0 < s < S_N = T - \tau$. Conditional on L , the reversed deletion times are the order statistics of L independent $\text{Uniform}(0, S_N)$ variables. Equivalently, when $N(X) < L$, the next-birth hazard is

$$B_L(X, s) = \frac{L - N(X)}{S_N - s}, \quad 0 \leq s < S_N, \quad (65)$$

and it is zero when $N(X) = L$. The apparent deadline singularity is benign: there are exactly $L \leq N_{\max}$ births, all strictly before S_N almost surely. The state follows the growth flow between them. Conditional on a birth, its mass and feature have learned normalized density

$$A_\theta^{(m)}(m, z \mid X, s, L), \quad 0 < m < v. \quad (66)$$

The timing rule is fixed; only the leaf-count law and birth marks are learned. Marginalizing L gives a valid latent-mixture birth process, but it need not be Markov in X alone because the posterior law of L can depend on the past. Keeping (X, L) as the augmented state makes the process Markov and makes exact birth-count support and non-explosion explicit.

Stage C: coagulation, $T - \tau < s < T$. The growth flow continues. Each unordered pair $a < b$ has merge intensity

$$K_\theta(a, b \mid X, s) \geq 0, \quad (67)$$

and the total event rate is

$$K_\theta^{\text{tot}}(X, s) = \sum_{a < b} K_\theta(a, b \mid X, s). \quad (68)$$

For a reversed training path, use the càdlàg convention $\tilde{X}_s = X_{(T-s)-}$ in the interior, with the endpoint conventions of (46). Let $\mathcal{E}_B$ and $\mathcal{E}_C$ be its birth and merge events. Its trainable negative log likelihood is

$$\begin{aligned} \mathcal{L}_{\text{path}}(\theta; H) = & -\log \Pi_\theta(L \mid M, y) - \sum_{e \in \mathcal{E}_B} \log A_\theta^{(m)}(m_e, z_e \mid \tilde{X}_{s_e^-}, s_e, L) \\ & + \int_{S_N}^T K_\theta^{\text{tot}}(\tilde{X}_s, s) ds - \sum_{e \in \mathcal{E}_C} \log K_\theta(a_e, b_e \mid \tilde{X}_{s_e^-}, s_e). \end{aligned} \quad (69)$$

The ordered birth times have fixed density $L!/S_N^L$; its hazard and survival terms do not depend on θ and are omitted from (69). Flow Jacobians and prescribed forward terms are also fixed in θ and appear only in the numerical ELBO.

The merge survival integral is essential. Training only observed pair labels would learn which merge occurred but not its timing or stopping law.

7.4 Unbiased integral estimator

For any interval $[a, b]$ and path-dependent rate $\Lambda_\theta(s)$,

$$\int_a^b \Lambda_\theta(s) ds = (b - a) \mathbb{E}_{U \sim \text{Unif}(a, b)} [\Lambda_\theta(U)]. \quad (70)$$

Thus one or more uniform times give an unbiased estimator of the merge survival integral. If the merge stage is divided into event-free intervals $I_j = [a_j, b_j]$, a one-sample stratified estimator is

$$\sum_j |I_j| K_\theta^{\text{tot}}(\tilde{X}_{U_j}, U_j), \quad U_j \sim \text{Uniform}(I_j). \quad (71)$$

The interval-length factors are required for unbiasedness.

7.5 Population consistency

The following theorem states the model-class condition behind the consistency claim.

Theorem 7.3 (Fisher consistency of factorized reverse learning). *Let Π^* , $A^{*(m)}$, and K^* be the exact leaf-count law, conditional birth-mark densities, and merge intensities of the augmented reverse process. Assume joint realizability: there is one parameter value θ^* for which all three model factors equal these exact factors simultaneously. Also assume that the learned factors assign positive density or intensity wherever the exact factors are positive, and that the displayed cross-entropies and compensator integrals are finite. Then the population risk of (69) is minimized by $\Pi_\theta = \Pi^*$, $A_\theta^{(m)} = A^{*(m)}$, and $K_\theta = K^*$ almost everywhere under the exact reverse path law.*

Proof. Subtract the risk at the exact factors. The leaf-count term becomes the expected categorical divergence $\mathbb{E}[\text{KL}(\Pi^* \parallel \Pi_\theta)]$. Conditional on every birth pre-state, time, and L , the birth-mark term becomes $\text{KL}(A^{*(m)} \parallel A_\theta^{(m)})$; summing over births and taking expectation keeps it nonnegative. For the merge process, the compensator identity turns the excess risk into

$$\mathbb{E} \int_{S_N}^T \sum_{a < b} \left[K_\theta - K^* + K^* \log \frac{K^*}{K_\theta} \right] ds. \quad (72)$$

For positive a, b , $a - b + b \log(b/a) \geq 0$, with equality only at $a = b$. We use the standard conventions $0 \log(0/a) = 0$ and assign infinite divergence when $b > 0$ but $a = 0$. Every term is therefore nonnegative, and all vanish exactly at the stated jointly realizable factors. $\square$

The theorem is conditional on joint realizability; it does not claim that a finite network recovers the exact factors from finite data. In particular, the bounded-rate implementation below may exclude an exact bridge whose rate diverges near the endpoint. The theorem does show why no explicit estimate of the marginal density ratios in (51)–(53) is required.

8 Neural parameterization

8.1 Permutation-equivariant encoder

Let $M_{\text{ref}} > 0$ be a fixed reference mass in the same units as M , and write $\ell_M = \log(M/M_{\text{ref}})$. This gives the networks a dimensionless total-mass input when examples have different M . For every component, form an embedding

$$h_i = \phi_\theta \left(z_i, \frac{m_i}{M}, s, \ell_M, y \right), \quad (73)$$

where y denotes optional conditioning information. A Deep Sets or Set Transformer backbone produces updated embeddings and a pooled summary

$$(\hat{h}_1, \dots, \hat{h}_n, g) = \text{SetNet}_\theta(h_1, \dots, h_n). \quad (74)$$

The component outputs are permutation equivariant and g is invariant [18, 13]. For the empty set, define $g = g_\emptyset$ using a learned null embedding (or a fixed zero vector followed by a network that also receives $v/M, s, \ell_M, L, y$). This makes the first birth-mark distribution in Algorithm 3 well defined.

8.2 Leaf count and birth marks

The leaf-count head is a categorical distribution on a finite support:

$$\Pi_\theta(\ell \mid M, y) = \text{softmax}(a_\theta(\ell_M, y))_\ell, \quad \ell = 0, \dots, N_{\text{max}}. \quad (75)$$

For theoretical statements, the categorical family includes boundary points with exact zero probabilities. In code, a masked softmax enforces known structural zeros; an ordinary softmax represents the interior and approaches an unknown zero only in the limit.

Write the birth mass as a fraction $\omega = m/v \in (0, 1)$. A fully normalized, directly implementable mark head is a mixture

$$A_\theta^{(\omega)}(\omega, z \mid X, s, L) = \sum_{j=1}^{J_{\text{mix}}} \alpha_j^{\text{mix}} \text{Beta}(\omega; \alpha_j, \beta_j) \mathcal{N}(z; \bar{z}_j, \text{diag}(\sigma_j^2)), \quad (76)$$

where all parameters depend on $(g, v/M, s, \ell_M, L, y)$, the mixture weights α_j^{mix} use a softmax, and positive parameters use softplus. The density with respect to $dm dz$ is

$$A_\theta^{(m)}(m, z \mid X, s, L) = \frac{1}{v} A_\theta^{(\omega)}(m/v, z \mid X, s, L). \quad (77)$$

For high-dimensional features, the Gaussian mixture may be replaced by a conditional normalizing flow, provided its density and samples are tractable.

8.3 Merge intensity

For every candidate unordered pair, compute a symmetric logit

$$\ell_{ab} = \psi_\theta(\hat{h}_a + \hat{h}_b, |\hat{h}_a - \hat{h}_b|, g, s, \ell_M, y) = \ell_{ba}. \quad (78)$$

When $N(X) \geq 2$, let

$$\pi_\theta(a, b \mid X, s) = \frac{\exp(\ell_{ab})}{\sum_{i < j} \exp(\ell_{ij})} \quad (79)$$

and parameterize a finite total rate with bounded average rate per pair,

$$\Lambda_\theta^C(X, s) = k_{\max}(s) \binom{N(X)}{2} \sigma(g_\theta(g, s, \ell_M, y)). \quad (80)$$

When $N(X) < 2$, set $\Lambda_\theta^C = 0$ and do not evaluate the pair softmax. Then

$$K_\theta(a, b \mid X, s) = \Lambda_\theta^C(X, s) \pi_\theta(a, b \mid X, s). \quad (81)$$

Because π_θ is normalized over the active candidate set, $K_\theta^{\text{tot}} = \Lambda_\theta^C$. The scaling in (80) permits a bounded average rate per candidate pair. Merge events cannot cause cardinality explosion because each one removes a component, although evaluating all logits still costs $O(N^2)$.

The measure-level reverse kernel is exact, but its coordinate merge rate can be unbounded, for example near $s = T$ under a hard endpoint-cardinality restriction or as a parent mass approaches zero when the density ratio does not cancel the $1/m$ factor. A finite $k_{\max}$ is then a regularized implementation and need not contain the exact bridge. The consistency theorem applies only under its stated realizability assumption. If exact hard final-count support is required, one may add a target-count bridge; we leave that extra conditioning variable out of the present model.

For large sets, construct a symmetric candidate graph $\mathcal{C}(X)$ using k nearest neighbors or learned locality, and normalize (79) only over $(a, b) \in \mathcal{C}(X)$. In this sparse mode replace (80) by

$$\Lambda_\theta^C(X, s) = k_{\max}(s) |\mathcal{C}(X)| \sigma(g_\theta(g, s, \ell_M, y)). \quad (82)$$

This preserves the bounded average rate per scored candidate and reduces pair scoring to $O(|\mathcal{C}|) = O(Nk)$. It is an approximation unless the true reverse assigns zero rate to omitted pairs. During likelihood training, every observed sibling pair must have nonzero model support. This can be enforced by using all pairs, by constructing an equivariant candidate graph with guaranteed support, or by mixing a sparse proposal with a positive full-pair fallback. Merely adding the target pair only during training would define a different model and is not valid.

Theorem 8.1 (Permutation invariance of the generated law). *Suppose the set encoder is permutation equivariant, the pooled summary is invariant, the pair logit is symmetric, and the leaf-count and birth-mark heads depend only on invariant summaries. In sparse mode, also suppose the candidate construction and its tie-breaking are permutation equivariant and retain the stated support. Starting from $\emptyset_M$, the distribution of the reverse process is invariant under every relabeling of component indices at every time.*

Proof. The empty state and sampled scalar L are invariant. Assume the current law is invariant. The fixed birth timing rule and birth-mark law are unchanged by relabeling, so a birth produces the same unordered law. Merge logits permute in the same way as their pairs, while the softmax denominator is invariant; hence pair selection is equivariant. The merge map itself is symmetric in its two inputs. The deterministic flow acts identically on every component. Induction over the finite event sequence proves invariance at every time. $\square$

Proposition 8.2 (Non-explosion of the learned sampler). *Assume $k_{\max}$ is bounded on $[S_N, T]$. For every sampled $L \leq N_{\max}$, the reverse process has at most L births and $\max\{L - 1, 0\}$ merges, hence at most $2N_{\max} - 1$ jumps.*

Proof. The conditional timing rule produces exactly L births. Stage C contains no births, and each merge lowers cardinality by one, so it can contain at most $\max\{L - 1, 0\}$ merges. The bound follows. Boundedness of $k_{\max}$ ensures every waiting-time calculation before the finite merge count is well defined. $\square$

9 Algorithms

9.1 Closed-form flow update

We use the following routine in all algorithms. For a forward segment from t_0 to t_1 , compute

$$G = \int_{t_0}^{t_1} \gamma(u) du, \quad r_1 = r_0 e^{-G}, \quad c_1 = M \frac{r_1}{1 + r_1}, \quad (83)$$

where $r_0 = c_0/(M - c_0)$. Set $m_i \leftarrow m_i(c_1/c_0)$ and $v \leftarrow M - c_1$. For a reverse segment from s_0 to s_1 , use

$$G_R(s_0, s_1) = \int_{s_0}^{s_1} \gamma(T - u) du = \int_{T-s_1}^{T-s_0} \gamma(t) dt. \quad (84)$$

Then replace e^{-G} by e^{+G_R} . Both updates preserve total mass up to floating-point roundoff. Setting $v = M - \sum_i m_i$ after each update removes accumulated roundoff. Before evaluating the ratio, handle the two boundary cases. If $c_0 = 0$, return the unchanged empty state. If $v_0 = 0$, return the unchanged state. In both cases the flow vector field is zero, and these branches avoid 0/0 or an infinite odds ratio.

Algorithm 1: Exact forward path simulation

1. **Input:** data state X_0 , horizon T , switch time τ , split rate β , split density κ , evaporation schedule γ .
2. Set $t \leftarrow 0$, $X \leftarrow X_0$, and initialize an empty event record. Give every component a persistent lineage identifier using the canonical convention of Section 7.
3. **Fragmentation stage:** while $t < \tau$:
 - (a) If $N(X) = 0$ or $N(X) = N_{\max}$, apply the forward flow to τ , set $t \leftarrow \tau$, and stop this stage.
 - (b) Draw $E \sim \text{Exp}(1)$ and seek the smallest $t' > t$ satisfying

$$\int_t^{t'} N(X) \beta(u) du = E.$$

If no solution exists before τ , apply the flow to τ , set $t \leftarrow \tau$, and stop this stage. For constant $\beta > 0$, this is $t' = t + E/[N(X)\beta]$; if $\beta = 0$, no solution exists.
 - (c) Apply the flow from t to t' and set $t \leftarrow t'$.
 - (d) Sample a parent uniformly and sample $(\rho, u) \sim \kappa_t$.
 - (e) Apply $X \leftarrow \mathcal{S}_i(X; \rho, u)$. Record the time, parent identifier, child identifiers, and split mark.
4. Set $t \leftarrow \tau$ if the fragmentation loop ended at the stage boundary.
5. **Deletion stage:** for every component alive at τ , independently draw $D_i \sim \text{Uniform}(\tau, T)$. Sort the deletion times.
6. Traverse the sorted times. Before each deletion, apply the forward flow from t to D_i , set $t \leftarrow D_i$, record the component state (m_i, z_i) immediately before deletion, and apply $X \leftarrow \mathcal{D}_i(X)$.
7. Apply the flow from t to T if needed, set $t \leftarrow T$, and set $X_T = \emptyset_M$.
8. **Return:** the event record and the reversible flow segments.

The deletion times can be sampled independently because Stage E contains no splits and every component has the same hazard (41). Sorting them is equivalent to simulating competing deletion clocks.

Algorithm 2: One stochastic training step

1. Sample a minibatch of data. For normalized raw weights, also draw $\xi \sim q_e$ and form the augmented state X_0 using (11). Simulate one forward history for each X_0 using Algorithm 1.
2. Reverse each history: reverse the event order, turn deletions into births with their recorded (m, z) marks, and turn splits into merges of the recorded sibling lineages.
3. Record the leaf budget $L = N(X_\tau)$ and evaluate:
 - the leaf-count term $\log \Pi_\theta(L \mid M, y)$;
 - for each birth, the absolute-mass log density

$$\log A_\theta^{(m)}(m, z \mid X, s, L) = \log A_\theta^{(\omega)}(m/v, z \mid X, s, L) - \log v;$$

- for each merge, $\log \Lambda_\theta^C(X, s) + \log \pi_\theta(a, b \mid X, s)$.

In sparse mode, enforce the nonzero-support requirement stated after (82); otherwise an observed merge can have infinite negative log likelihood.

4. Estimate the merge survival integral in (69). Either sample uniform times as in (70), or use the interval-length-weighted stratification in (71). Birth times have a fixed law and require no learned survival term.
 5. Average (69) over the minibatch, differentiate it, and update θ .
 6. If a numerical ELBO is wanted, also accumulate the prescribed forward density, joint Jacobian, fixed birth-time density, and any dequantization terms. None is needed for the gradient with respect to θ .
-

Algorithm 3: Event-driven reverse sampling by thinning

1. **Input:** trained model, M , condition y , horizon T , switch time τ , and valid upper envelopes for merge rates.
 2. Set $s \leftarrow 0$ and $X \leftarrow \emptyset_M$. Sample $L \sim \Pi_\theta(\cdot \mid M, y)$.
 3. **Nucleation stage**, $0 < s < T - \tau$:
 - (a) If $L = 0$, apply reverse growth from 0 to $T - \tau$ (the empty state remains unchanged), set $s \leftarrow T - \tau$, skip the remaining nucleation substeps, and continue to the coagulation stage.
 - (b) Draw L independent times from $\text{Uniform}(0, T - \tau)$ and sort them as $0 < s_1 < \dots < s_L < T - \tau$.
 - (c) For $j = 1, \dots, L$, apply reverse growth from the current time to s_j , set $s \leftarrow s_j$, sample (ω, z) from (76), set $m = \omega v$, and apply $X \leftarrow \mathcal{B}_{m,z}(X)$.
 - (d) After the last birth, apply reverse growth to $T - \tau$ and set $s \leftarrow T - \tau$.
 4. **Coagulation stage**, $T - \tau < s < T$:
 - (a) If $N(X) < 2$, apply reverse growth to T and stop. If a sparse candidate set is empty, fall back to the full pair set by default, or advance only to the next scheduled candidate-refresh time. Stop early only if emptiness is known to remain valid until T .
 - (b) Choose an interval $I = [s, s_I]$ and a valid upper envelope $\bar{K}$ for Λ_θ^C along its deterministic flow. For full pair scoring, a direct choice is

$$\bar{K} = \binom{N(X)}{2} \sup_{u \in I} k_{\max}(u).$$

In sparse mode, replace the binomial factor by a valid upper bound on $|\mathcal{C}(X_u)|$ over $u \in I$. If $\bar{K} = 0$, flow to s_I , set $s \leftarrow s_I$, and recompute; return to the coagulation-stage test, and stop only when $s_I = T$.
 - (c) Draw $E_K \sim \text{Exp}(\bar{K})$. If $s + E_K \geq s_I$, flow to s_I , set $s \leftarrow s_I$, and return to the coagulation-stage test. Otherwise flow to $s + E_K$ and set $s \leftarrow s + E_K$.
 - (d) Accept with probability $\Lambda_\theta^C(X, s)/\bar{K}$. On acceptance, sample an unordered pair from $\pi_\theta(a, b \mid X, s)$ and apply $X \leftarrow \mathcal{C}_{ab}(X)$. Whether accepted or rejected, recompute or revalidate the next envelope and repeat the coagulation-stage test.
 5. Return the unordered component set and report the remaining reservoir v . For a nonempty normalized-weight output, decode $w_i = m_i / \sum_j m_j$; treat the empty set as a separate outcome.
-

The thinning algorithm assumes a finite envelope on each interval. If an unbounded bridge schedule is deliberately used near T , simulate it through its integrated hazard or a time change; a single finite envelope extending to T does not exist.

When analytic envelopes are inconvenient, a time grid may replace merge thinning. At the left endpoint (X, s) , evaluate all merge rates and set $\Lambda = K_\theta^{\text{tot}}(X, s)$. Fix a step-size tolerance $\rho_{\text{grid}} \in (0, 1)$ and choose $\Delta \leq T - s$. When $\Lambda > 0$, also require $\Lambda\Delta \leq \rho_{\text{grid}}$ and use

$$p_{\text{event}} = 1 - e^{-\Lambda\Delta}. \quad (85)$$

Draw a Bernoulli variable with this probability. If it is one, draw and retain a pair from $K_\theta(a, b \mid X, s)/\Lambda$. Apply exact reverse growth from s to $s + \Delta$, set $s \leftarrow s + \Delta$, and then merge the retained pair if an event was drawn. Thus rates and pair probabilities are frozen at the left endpoint and at most one event occurs per step. This is a first-order approximation and converges as $\max \Delta \rightarrow 0$. If a multiple-event tau-leap is used instead, events must be applied sequentially because each merge changes both the available pairs and their rates. The grid changes timing accuracy but not mass conservation. When $\Lambda = 0$, set $p_{\text{event}} = 0$, apply only the growth update over Δ , and do not evaluate K_θ/Λ .

9.2 Computational complexity

Let $C_{\text{birth}}(N)$ be the cost of one birth-mark evaluation at cardinality N , and let $C_{\text{enc}}(N)$ be the cost of one set-encoder evaluation. Let N_ℓ be the active cardinality at birth evaluation ℓ . Let P_C be the number of merge-network evaluations, including accepted events, rejected thinning proposals, and survival-integral

queries, and let N_j be the active cardinality at its j th evaluation. Sorting birth times costs $O(L \log L)$. With full pair scoring, the total sampling and rate-evaluation cost is

$$O\left(L \log L + \sum_{\ell=1}^L C_{\text{birth}}(N_\ell) + \sum_{j=1}^{P_C} [C_{\text{enc}}(N_j) + N_j^2]\right). \quad (86)$$

A sparse k -neighbor candidate graph replaces N_j^2 by $|\mathcal{C}(X_j)| = O(N_j k)$. Batched survival evaluations dominate training when many quadrature times are used; caching the fixed reversed path states and batching all queried times reduces overhead.

10 Further theoretical properties

10.1 Endpoint error is controlled by path error

Let $\mathbb{P}^*$ be the exact augmented reverse path law obtained from Theorem 6.3, and let $\mathbb{P}_\theta$ be the learned reverse path law. Both begin at $\mathcal{O}_M$ and first sample their respective leaf budgets. Write P_{data} and P_θ^T for their endpoint probability measures. These are induced by the augmented data law and learned endpoint densities, respectively.

Theorem 10.1 (Path-to-endpoint error bound). *If $\mathbb{P}^* \ll \mathbb{P}_\theta$, then*

$$\text{KL}(P_{\text{data}} \parallel P_\theta^T) \leq \text{KL}(\mathbb{P}^* \parallel \mathbb{P}_\theta). \quad (87)$$

Consequently,

$$\text{TV}(P_{\text{data}}, P_\theta^T) \leq \sqrt{\frac{1}{2} \text{KL}(\mathbb{P}^* \parallel \mathbb{P}_\theta)}. \quad (88)$$

Proof. The endpoint is a measurable function of the full path. Relative entropy cannot increase under a measurable map, which proves (87). Equation (88) is Pinsker’s inequality. $\square$

For normalized observations, applying the decoder (13) is another measurable map. The same data-processing argument therefore also bounds the mismatch between the decoded nonempty endpoint laws, with the empty outcome retained as its separate atom.

When the exact and learned laws use the same deterministic flow, the same conditional birth-time law, a common record reference measure and support, and integrable intensities, the path-space KL decomposes into the leaf-count KL, conditional birth-mark KL terms, and the merge-intensity divergence in (72). Equivalently, the *population excess* of the full reverse-record negative log likelihood over its value at the exact reverse law equals this path KL. It is this excess risk, not the raw nonnegative loss value, that upper-bounds the endpoint KL through Theorem 10.1. If the flow, timing law, or control law is also learned, its corresponding path-space term must be added; incompatible supports give infinite KL.

10.2 Hard conservation is stronger than a penalty

Corollary 10.2 (Zero structural conservation error). *In exact arithmetic, every sample returned by Algorithm 3 satisfies*

$$v + \sum_i m_i = M. \quad (89)$$

This statement holds for every parameter value θ , not only at an optimum.

Proof. Algorithm 3 is a composition of the reverse growth flow, birth maps, and merge maps. Proposition 4.1 shows that each map is conservative. $\square$

This is why the proposed method does not include a mass-conservation loss. A penalty can make violations small on average; it cannot provide the above structural guarantee.

10.3 Conditioning on varying total mass

If total mass varies across observations, first model

$$M \sim p_\vartheta(M \mid y) \quad (90)$$

with a one-dimensional density, or treat M as observed conditioning information. Conditional on the sampled or supplied M , the entire coagulation–fragmentation process remains conservative. Mixing examples with different M without either step would make the claimed conservation law ill-defined. The parameterization above supplies $\ell_M = \log(M/M_{\text{ref}})$ to the encoder, birth, pair, and total-rate heads. Omitting that input would restrict the model to scale-invariant conditional laws even if M were sampled separately. In the first case, training adds

$$\mathcal{L}_M = -\log p_\vartheta(M \mid y), \quad (91)$$

and Algorithm 3 samples $M \sim p_\vartheta(\cdot \mid y)$ before initializing $\mathcal{O}_M$. In the experiments proposed here, we instead condition on or normalize M so that this extra head is not required.

10.4 Feature diffusion is outside the present model

All theorems, likelihoods, and algorithms in this paper assume deterministic feature motion between jumps. Adding Brownian feature noise would require a reverse-score or Girsanov term and an SDE solver. That is a possible future extension, but it is not part of the model analyzed here.

11 Extension: locally controlled evaporation

11.1 What the hair-dryer observation adds

The mirror episode suggests more than a global evaporation schedule. A hair dryer clears a small region, the user sees which regions remain covered, and the dryer is moved again. The useful abstract properties are therefore *locality*, *sequential action*, and *feedback*. This section adds those properties without changing the split, merge, deletion, or birth maps of the core model.

There is also an important distinction. On a real mirror, the reflected face already exists behind the droplets. Literal mirror clearing is thus a restoration or de-occlusion problem, not unconditional generation. We use the hair dryer as motivation for a spatial control mechanism. We do not claim that the hidden face and the generated weighted set are the same mathematical object.

11.2 A spatial control field

Some coordinates of z_i need not be spatial. Let $x_i^{\text{sp}} = \text{pos}(z_i) \in \mathbb{R}^{q_s}$ be a specified spatial projection, and let $a_t : \mathbb{R}^{q_s} \rightarrow [0, 1]$ describe where the dryer acts at forward time t . A convenient finite representation uses K_{basis} nonnegative spatial basis functions satisfying $\sum_{k=1}^{K_{\text{basis}}} \psi_k(x) \leq 1$:

$$a_t(x) = \sum_{k=1}^{K_{\text{basis}}} a_{t,k} \psi_k(x), \quad a_{t,k} \in [0, 1]. \quad (92)$$

The functions may be image patches, radial kernels, or cells of a spatial grid. The coefficient vector determines the location and strength of the current action.

For component i , let

$$\lambda_i(t, X, a_t) = \lambda_0(t) + \lambda_1(t) a_t(x_i^{\text{sp}}) \geq 0, \quad (93)$$

where $\lambda_0(t), \lambda_1(t) \geq 0$, or use any bounded nonnegative permutation-equivariant rate. Replace the global flow by

$$\frac{dm_i}{dt} = -\lambda_i(t, X_t, a_t) \frac{v}{M} m_i, \quad (94)$$

$$\frac{dv}{dt} = \frac{v}{M} \sum_{i=1}^{N(X_t)} \lambda_i(t, X_t, a_t) m_i, \quad (95)$$

$$\frac{dz_i}{dt} = 0. \quad (96)$$

The original model is recovered by setting $\lambda_i(t, X, a) = \gamma(t)$ for every component. Unlike the global flow, the controlled flow generally changes relative component masses and does not reduce to one scalar odds equation.

Proposition 11.1 (Conservation and positivity under local control). *Assume the action path is piecewise continuous. Assume the rates λ_i are bounded, nonnegative, and uniformly Lipschitz in the state on the compact mass simplex, uniformly over time and admissible actions. Then the controlled ODE has a unique solution on every finite time interval. If initially $m_i \geq 0$, $v \geq 0$, and $v + \sum_i m_i = M$, these properties hold for all later times. If the rates are permutation equivariant, so is the flow.*

Proof. On each interval where the action is continuous, the regularity assumptions give a unique local solution; concatenating the intervals handles action jumps. The state remains in the compact mass simplex, so the bounded vector field cannot explode and the solution extends through every finite interval. At $m_i = 0$, the right side of (94) is zero. At $v = 0$, all mass derivatives in (94)–(95) are zero. Thus no coordinate can cross below zero. Finally,

$$\frac{d}{dt} \left(v + \sum_i m_i \right) = \frac{v}{M} \sum_i \lambda_i m_i - \frac{v}{M} \sum_i \lambda_i m_i = 0.$$

The total is therefore M . Relabeling the components only relabels an equivariant vector field, which proves permutation equivariance. $\square$

11.3 Conditional time reversal

The cleanest exact construction treats the complete action path $a_{0:T}$ as known conditioning information. Let P_t^a and Q_t^a denote the forward marginal and jump kernel conditional on this path. The reverse jumps still satisfy the same flux identity,

$$P_t^a(dY) \bar{Q}_{T-t}^a(Y, dX) = P_t^a(dX) Q_t^a(X, dY). \quad (97)$$

Between jumps, the reverse traverses the controlled ODE backward:

$$\frac{dm_i}{ds} = +\lambda_i(T-s, Y_s, a_{T-s}) \frac{v}{M} m_i, \quad (98)$$

$$\frac{dv}{ds} = -\frac{v}{M} \sum_i \lambda_i(T-s, Y_s, a_{T-s}) m_i. \quad (99)$$

Thus the reverse process performs stronger growth in the spatial regions that were evaporated more strongly in the forward direction.

The closed-form flow Jacobian (132) must be replaced by an ODE Jacobian. Let $b^a(m, t)$ be the vector formed by the right sides of (94), with $v = M - \sum_i m_i$. For a no-jump segment from r to t , additionally assume that $b^a(\cdot, t)$ is C^1 on the mass-simplex interior and has integrable divergence along the trajectory. Liouville's formula then gives

$$\log |\det D\Phi_{r,t}^a(m_r)| = \int_r^t \nabla_m \cdot b^a(m_u, u) du. \quad (100)$$

Equivalently, the multiplicative flow factor used in a change of variables is

$$J_{\text{flow}}^a(r, t) = \exp \left\{ \int_r^t \nabla_m \cdot b^a(m_u, u) du \right\}. \quad (101)$$

The log factor can be integrated together with the state. When each λ_i depends on $(t, x_i^{\text{sp}}, a_t)$ but not on the masses,

$$\nabla_m \cdot b^a(m, t) = \frac{1}{M} \sum_i \lambda_i(t, x_i^{\text{sp}}, a_t)(m_i - v). \quad (102)$$

For a mass-dependent neural rate, automatic differentiation gives the full divergence. Equation (102) treats the action on each segment as fixed conditioning information. If the action is recomputed differentially from the current state, the divergence must also differentiate through that controller.

The reverse merge-rate factor $1/m$ in (53) is unchanged. Separately, every forward split still contributes its parent mass m to the joint-history Jacobian in (133). Each global-flow factor in that Jacobian is replaced by J_{flow}^a from (101). For one prescribed control path, the algebraic form of the trainable jump loss (69) is unchanged, but A_θ and K_θ must condition on that path, or on a sufficient current control-state encoding. If the control or controlled-flow parameters are learned jointly with the generator, the action density and controlled-flow Jacobian are parameter-dependent and must also be included in the objective.

11.4 Feedback and adaptive stopping

An open-loop controller fixes $a_{0:T}$ before the process starts. The mirror story instead suggests a closed-loop controller. At decision time t_k , let o_k be an observation or uncertainty summary and sample

$$a_k \sim \pi_\varphi(\cdot \mid X_{t_k}, o_k, t_k, h_k^{\text{ctrl}}), \quad (103)$$

where h_k^{ctrl} is optional controller memory. We take o_k and the memory update to be deterministic functions of the augmented state and past actions. If either is stochastic, its transition density must also be included in the history likelihood. The sampled coefficient vector is held fixed on $[t_k, t_{k+1})$, or interpolated by a stated deterministic rule, to define the continuous field a_t used by the ODE.

The process is Markov after augmenting its state with $(o_k, h_k^{\text{ctrl}}, a_k)$. There is one extra point for fixed decision times: these are predictable transitions and are not covered by the no-predictable-jump condition in Assumption 6.2. Let Z_k^- and Z_k^+ be the augmented states immediately before and after decision t_k , and let $R_k(Z_k^-, dZ_k^+)$ be the forward scheduled-transition kernel. Its pre- and post-decision laws are denoted by $P_{t_k}^-$ and P_{t_k} . Its exact reverse is the discrete Bayes kernel $\bar{R}_k$ defined by

$$P_{t_k}(dZ^+) \bar{R}_k(Z^+, dZ^-) = P_{t_k}^-(dZ^-) R_k(Z^-, dZ^+). \quad (104)$$

Between decision times, the continuous flow and ordinary jumps use (97). Hazard-driven random decision times can instead be included in the ordinary augmented jump kernel. Noninvertible memory updates cause no problem for (104), but the forward history must retain the predecessor state and any sampled noise used to train its reverse.

Let J denote the number of control decisions in a realized trajectory. Three probability conventions must be kept separate:

1. A prescribed action path is conditioning information and contributes no action-density term.
2. If π_φ is the forward controller used in the variational corruption law, then

$$\log q_\varphi(a_{1:J}, H \mid X) \supset \sum_k \log \pi_\varphi(a_k \mid X_{t_k}, o_k, t_k, h_k^{\text{ctrl}}), \quad (105)$$

and this term enters the ELBO with a minus sign through $-\log q_\varphi$.

3. If actions are latent variables of the generative model, parameterize a reverse scheduled kernel $\bar{R}_{\omega,k}$ satisfying

$$Z_k^- \sim \bar{R}_{\omega,k}(Z_k^+, dZ_k^-),$$

or use a hazard-driven reverse action kernel. Its eventwise log density $\sum_k \log \bar{r}_{\omega,k}(Z_k^- \mid Z_k^+)$ enters the generative path likelihood with a plus sign.

An online forward controller cannot be used during generation merely by “reversing its actions,” because no realized forward action path exists at sampling time. Its action transitions must themselves be reversed through the augmented-state flux identity or modeled by the separate reverse kernel in item 3.

Let $U_\psi(o_k) \geq 0$ measure how much of the target remains unresolved. A simple stopping time is

$$\tau_{\text{stop}} = \min\{T, \inf\{t_k : U_\psi(o_k) \leq \epsilon\}\}, \quad (106)$$

where $\inf \emptyset = +\infty$. The controller can be learned from action demonstrations using the negative of (105). It may instead be optimized with the cost

$$\mathcal{J}_{\text{ctrl}}(\varphi) = \mathbb{E} \left[U_\psi(o_J) + \lambda_a \sum_{k=1}^J \text{Cost}(a_k) + \lambda_{\text{step}} J \right]. \quad (107)$$

The first term rewards revealing the target, the second penalizes dryer energy or spatial coverage, and the third penalizes long action sequences. This feedback objective is useful for conditional restoration or inpainting. For unconditional generation, the fixed horizon T remains the simpler and better-defined stopping rule. Equation (106) therefore defines a separate control or restoration objective; it is not silently part of the fixed-horizon generative model. If adaptive stopping is used inside that model, the stopping decision needs an explicit probability or hazard and a random terminal law. Both must appear in the forward variational and reverse generative path densities.

11.5 Implementation

Algorithm 4: Controlled-flow modification

1. Choose spatial basis functions $\psi_{1:K_{\text{basis}}}$. Select one action convention: a prescribed path; a forward variational controller together with a learned reverse action kernel; or a generative action prior with its reverse controller.
 2. In Algorithm 1, keep the same split and deletion events. Between events, integrate (94)–(95) and, when a numerical ELBO is required, simultaneously integrate the divergence in (100). Record every action, observation, and controller-memory state used by the forward law.
 3. For a prescribed path, reverse its time ordering and use a_{T-s} . For latent or online actions at fixed decision times, train and sample $\bar{R}_{\omega,k}$ using (104), and add $\sum_k \log \bar{r}_{\omega,k}$ to the reverse-record log likelihood. For hazard-driven decisions, use the augmented continuous-time reverse kernel. Do not replay a forward action path that does not exist at generation time.
 4. In Algorithm 3, replace each closed-form growth update by numerical integration of (98)–(99). Birth and merge sampling are otherwise unchanged.
 5. A prescribed path adds no action-density term. A forward variational controller contributes through $-\log q_\varphi$, whereas a generative action prior or reverse policy contributes through $+\log p$. Use (106) only for a modeled conditional revealing task; otherwise retain the fixed horizon T .
-

Use a positivity-preserving, mass-conserving ODE integrator with step rejection. Recomputing $v = M - \sum_i m_i$ solely to remove floating-point roundoff is harmless. Material clipping or projection, however, changes the numerical map and invalidates the continuous Liouville Jacobian in (100). If such a discrete solver is used, the implementation must instead compute the Jacobian of its actual discrete map. The local model is slower than the scalar closed-form flow, so the global model should remain the default unless spatial control yields a measurable benefit.

11.6 Airflow, departure, and the modeled boundary

A physical hair dryer can also translate droplets or push them off the mirror. Deterministic translation of the spatial coordinates may be represented by $dx_i^{\text{sp}}/dt = u_{\text{air}}(t, x_i^{\text{sp}}, a_t)$. If water leaves the modeled region, transferring it directly to v would incorrectly call it local vapor. Introduce an escaped reservoir e instead and conserve

$$v + e + \sum_i m_i = M. \quad (108)$$

A departure event removes component i and adds m_i to e ; its reverse is an insertion drawing mass from e . These transport and departure events are not needed for the minimal local-evaporation extension and are left out of the main experiments.

11.7 Experiments specific to this extension

The controlled model should first be tested on synthetic spatially heterogeneous sets or a conditional de-occlusion task. Compare four matched controllers: global uniform action, a fixed raster scan, an uncertainty-based oracle, and a learned feedback policy. Report generation or restoration quality together with the action energy $\int_0^T \|a_t\|_1 dt$, number of control decisions, wall-clock time, and mass residual. The extension is useful only if it improves quality or reduces computation at a matched action budget.

12 Experimental plan (results pending)

Status. No experiment in this section has been run. The hypotheses, baselines, and metrics below are a prospective evaluation plan. Numerical tables are intentionally blank, and the text makes no empirical claim.

12.1 Research questions

H1: Reverse correctness.

On a finite process with exactly computable marginals, learned reverse rates approach the analytical rates.

H2: Benefit of coagulation.

At matched architecture and compute, pairwise merging captures pair and cluster statistics better than a single-component insertion model with no pair-merge operation.

Check 1: Conservation.

Verify as an implementation check that the sampler preserves total mass to numerical precision; this is a theorem, not an empirical hypothesis.

H3: Cardinality.

After sampling an auxiliary leaf budget and then merging leaves, the model matches the held-out *final* cardinality distribution. This does not claim that the method has no count model.

H4: Efficiency.

Sample quality as a function of wall-clock time, event count, and peak active cardinality is competitive with trans-dimensional baselines.

H5: Local control.

On a spatially heterogeneous conditional task, the extension in Section 11 improves quality or reduces action cost relative to uniform evaporation and a fixed scan.

12.2 Benchmarks

1. **Finite enumerated process.** Use integer masses, a small feature lattice, and small $N_{\max}$. Compute P_t and the exact reverse generator by solving the finite Kolmogorov equation. This tests probability-flux reversal and unordered pair counting, but not the continuous $1/m$ Jacobian.
2. **Continuous synthetic weighted sets.** Sample a known hierarchical fragmentation process with controllable split imbalance, spatial interaction strength, and cardinality. Retain the ground-truth genealogy only for evaluation. Separately test the continuous split change of variables by Monte Carlo integration to verify the $1/m$ factor.

3. **One real scalar-weighted set domain.** Use a dataset in which the additive scalar in the present theory is measured rather than invented. Two compatible candidates are calorimeter-hit point clouds with deposited energy as m_i , conditioned on total deposited energy, and instance-segmented droplet microscopy with calibrated droplet volume as m_i . Vector four-momentum conservation is outside the present scalar theory and is left to future work.

12.3 Baselines

- Jump Diffusion Models with the same set encoder [8].
- Existence-Field Diffusion on spatial point-process tasks [15].
- A fixed- or cardinality-conditioned diffusion baseline.
- An autoregressive split-tree model when a branching domain is used.
- The analytical reverse generator on the finite benchmark.
- **Critical in-house ablation:** a birth-only single-component insertion process using the same backbone, parameter count, training paths, and compute budget, but no pairwise merging.

12.4 Metrics

Report:

- reverse-rate relative error and event calibration on the exact task;
- birth-mark negative log likelihood and merge-pair accuracy;
- total variation or Hellinger distance between cardinality distributions;
- Wasserstein distance for component-mass marginals;
- pair-correlation or Ripley- K error for spatial structure;
- MMD, energy distance, and a classifier two-sample test on invariant set summaries;
- the relative conservation residual

$$\epsilon_M = \frac{|v + \sum_i m_i - M|}{M + 10^{-12}}; \quad (109)$$

- held-out ELBO on the finite task and on continuous tasks only after the joint Jacobian and dequantization terms have been implemented and checked; exact likelihood only where it is genuinely computable;
- for local control, integrated action energy, spatial coverage, control-decision count, and terminal unresolved uncertainty;
- wall-clock sampling time, neural evaluations, jump count, average and maximum active cardinality, and peak memory.

12.5 Ablations

The minimum ablation set is:

1. pairwise coagulation versus the birth-only in-house ablation;
2. exact conservation versus an unconstrained penalty-based model;
3. learned pair kernel versus uniform and distance-only kernels;
4. reservoir versus no explicit reservoir;
5. coupled versus factorized birth marks;
6. full pair enumeration versus sparse candidate pairs;

7. several fragmentation strengths and switch times τ ;
8. for the optional extension, global control versus a fixed local scan, an uncertainty oracle, and a learned feedback policy.

All models should use identical data splits and comparable encoder capacity. Report five independent random seeds with 95% bootstrap confidence intervals and include quality–compute curves rather than only one operating point.

12.6 Planned reporting format

Table 1: Finite reverse-process validation. Results pending.

Method	Rate error	Mark NLL	Cardinality TV
Analytical reverse	—	—	—
Birth-only in-house ablation	—	—	—
Proposed coagulation model	—	—	—

Table 2: Continuous weighted-set generation. Results pending.

Method	Set distance	Pair-stat. error	ϵ_M	Time/sample
Cardinality-conditioned model	—	—	—	—
Jump diffusion	—	—	—	—
Birth-only in-house ablation	—	—	—	—
Proposed coagulation model	—	—	—	—

13 Limitations

The method has several clear limitations. Unrestricted pair scoring is quadratic in active cardinality. Sparse candidates reduce cost but may remove important long-range merges. The finite cap $N_{\max}$ and the auxiliary leaf-count model are practical structural choices; a poor cap or count model can limit support. Simulated forward histories may also have high variance, so the fragmentation schedule is a modeling decision rather than a harmless implementation detail.

The scalar conservation law is useful only when m_i has a defensible additive meaning. Normalizing arbitrary scalar scores to sum to one would make the constraint cosmetic. The present split map preserves a Euclidean weighted centroid. Categorical attributes, molecules with typed atoms, and systems with vector-valued conservation laws require new event maps and mark models. Normalized data also require the reservoir dequantization and decoder described in Section 3.

The prescribed-path local-control extension has an exact conditional reversal. A feedback-controlled application additionally needs a task-specific observation model and the scheduled reverse kernels in (104); Algorithm 4 specifies where these terms enter but does not choose one universal controller architecture.

The exact terminal construction uses a dedicated deletion stage. This is mathematically clean but less physically literal than simultaneous nucleation, growth, and coagulation. The condensation language is therefore an intuition, not a claim that the algorithm implements Hertz–Knudsen kinetics or an equilibrium free energy. The general jump-flux identity and variable-dimensional generation are also prior ideas; the claimed technical contribution is their conservative unordered split–merge specialization and the resulting learning and sampling scheme.

Finally, a powerful birth-mark model could explain most of the data while merges add little. The birth-only in-house ablation is therefore necessary to establish that coagulation provides more than an attractive interpretation.

14 Conclusion

We developed a complete theoretical and algorithmic specification of a mass-conserving generative process on variable-size weighted sets. A prescribed fragmentation–evaporation process reaches an exact all-reservoir state. Its finite-horizon reverse performs nucleation, conservative growth, and pairwise coagulation. The reverse rates follow from probability flux, while a path-space ELBO turns simulated corruption histories into a tractable marked point-process loss. Conservation and permutation invariance hold by construction.

The remaining question is empirical rather than definitional: does pairwise coagulation improve modeling or efficiency compared with the birth-only ablation and existing variable-cardinality methods? Section 12 specifies the experiments needed to answer that question without presupposing the result.

A Derivation of the reservoir dequantization formulas

This section derives both the Jacobian in (12) and the likelihood bound in (14). Write

$$w_n = 1 - \sum_{i=1}^{n-1} w_i, \quad m_i = M(1 - \xi)w_i, \quad i = 1, \dots, n. \quad (110)$$

Thus $w_1, \dots, w_{n-1}$ are the independent simplex coordinates, while ξ determines the total amount of mass placed in the reservoir.

If ξ were fixed at ξ_0 , the image would satisfy the extra equality $\sum_i m_i = M(1 - \xi_0)$. It would therefore have only $n - 1$ mass coordinates inside an n -dimensional mass domain, and hence zero n -dimensional Lebesgue measure. Allowing ξ to vary supplies the missing direction. Conversely, for any nonempty interior state, $\xi = v/M$ and $w_i = m_i / \sum_j m_j$ recover its unique normalized coordinates, as proved in Proposition 3.1.

A.1 Jacobian determinant

For $n = 1$, we have $w_1 = 1$ and $\partial m_1 / \partial \xi = -M$, so the claimed absolute determinant is $M = M^1(1 - \xi)^0$. Assume $n \geq 2$ for the block calculation below. Define $a = M(1 - \xi)$ and $\tilde{w} = (w_1, \dots, w_{n-1})^\top$. For $i < n$,

$$\frac{\partial m_i}{\partial w_j} = a\delta_{ij}, \quad \frac{\partial m_i}{\partial \xi} = -Mw_i. \quad (111)$$

Because $m_n = a(1 - \sum_{i < n} w_i) = aw_n$, the final row satisfies

$$\frac{\partial m_n}{\partial w_j} = -a, \quad \frac{\partial m_n}{\partial \xi} = -Mw_n. \quad (112)$$

Therefore the full Jacobian matrix is

$$\frac{\partial(m_1, \dots, m_n)}{\partial(w_1, \dots, w_{n-1}, \xi)} = \begin{pmatrix} aI_{n-1} & -M\tilde{w} \\ -a\mathbf{1}^\top & -Mw_n \end{pmatrix}. \quad (113)$$

Using the block-determinant identity $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A) \det(D - CA^{-1}B)$ gives

$$\begin{aligned} \det J &= a^{n-1} [-Mw_n - (-a\mathbf{1}^\top)(aI_{n-1})^{-1}(-M\tilde{w})] \\ &= a^{n-1} \left[-Mw_n - M \sum_{i=1}^{n-1} w_i \right] \\ &= -Ma^{n-1} = -M^n(1 - \xi)^{n-1}. \end{aligned} \quad (114)$$

Taking the absolute value proves

$$|\det J| = M^n(1 - \xi)^{n-1}. \quad (115)$$

The exponent $n - 1$ appears because the normalized weight vector has $n - 1$ independent simplex directions, each scaled by $M(1 - \xi)$. The remaining radial direction is controlled by ξ and contributes the final factor M . The feature coordinates are unchanged and therefore contribute a Jacobian factor of one.

A.2 Raw normalized-data likelihood

For compact notation, define

$$f_\theta(\xi; w, z) := p_{\theta,n}(X(\xi))M^n(1-\xi)^{n-1}, \quad 0 < \xi < 1. \quad (116)$$

Here $p_{\theta,n}$ is the *joint* endpoint density of all masses and features relative to μ_n ; it is not a separately specified density for z . Under the change of variables above, the model density in coordinates $(w_{1:n-1}, z_{1:n}, \xi)$ is

$$\tilde{p}_\theta(w_{1:n-1}, z_{1:n}, \xi) = f_\theta(\xi; w, z). \quad (117)$$

The same unordered counting convention is used before and after this change of variables, so it introduces no additional factorial. Since ξ is not part of the raw observation, its likelihood is obtained by marginalization:

$$p_\theta^{\text{raw}}(w_{1:n-1}, z_{1:n}) = \int_0^1 f_\theta(\xi; w, z) d\xi. \quad (118)$$

This integral must cover all state-implied reservoir fractions. The auxiliary density q_ϵ is supported only on $I_\epsilon = [\epsilon_{\min}, \epsilon_{\max}]$, so define the restricted mass

$$Z_{I_\epsilon}(w, z) := \int_{I_\epsilon} f_\theta(\xi; w, z) d\xi. \quad (119)$$

Insert $q_\epsilon(\xi)/q_\epsilon(\xi)$ only on this interval. Then

$$p_\theta^{\text{raw}}(w_{1:n-1}, z_{1:n}) \geq Z_{I_\epsilon}(w, z) = \mathbb{E}_{\xi \sim q_\epsilon} \left[\frac{f_\theta(\xi; w, z)}{q_\epsilon(\xi)} \right]. \quad (120)$$

The first relation is an inequality unless the endpoint model has no conditional mass outside I_ϵ . Monotonicity of the logarithm and then Jensen's inequality yield

$$\log p_\theta^{\text{raw}}(w_{1:n-1}, z_{1:n}) \geq \mathbb{E}_{\xi \sim q_\epsilon} \left[\log p_{\theta,n}(X(\xi)) + n \log M + (n-1) \log(1-\xi) - \log q_\epsilon(\xi) \right], \quad (121)$$

which is (14).

Assume the displayed raw likelihood and $Z_{I_\epsilon}(w, z)$ are positive. The gap then has the usual variational interpretation. Define

$$p_\theta(\xi \mid w, z) = \frac{f_\theta(\xi; w, z)}{p_\theta^{\text{raw}}(w_{1:n-1}, z_{1:n})}. \quad (122)$$

Then the difference between the left side of (121) and its right side is

$$\text{KL}(q_\epsilon(\xi) \parallel p_\theta(\xi \mid w, z)) \geq 0. \quad (123)$$

Equivalently, the same gap can be separated as

$$\log \frac{p_\theta^{\text{raw}}(w, z)}{Z_{I_\epsilon}(w, z)} + \text{KL} \left(q_\epsilon(\xi) \parallel \frac{f_\theta(\xi; w, z)}{Z_{I_\epsilon}(w, z)} \right). \quad (124)$$

The bound is exact only when the full conditional posterior is supported on I_ϵ and equals q_ϵ almost everywhere.

B Full proof of the split counting factor

We give a more explicit version of the calculation behind Proposition 6.8. Hold the $n-1$ unaffected components fixed and call their coordinate element dO . Selecting one parent from an unordered n -component state changes the measure factor as

$$\frac{1}{n!} \sum_{i=1}^n dO \, dm \, dz = \frac{1}{(n-1)!} dO \, dm \, dz. \quad (125)$$

Adding the canonical split-mark element gives

$$\frac{1}{(n-1)!} dO dm dz d\rho du. \quad (126)$$

In the $(n+1)$ -component child state, select one physical unordered pair. The labeled-coordinate integral contains both label assignments for its two members, even though the sum below is written with $a < b$. Counting the candidate pairs gives

$$\frac{1}{(n+1)!} \sum_{a < b} dO dm_a dm_b dz_a dz_b = \frac{1}{2(n-1)!} dO dm_a dm_b dz_a dz_b. \quad (127)$$

Lemma 6.5 gives

$$dm_a dm_b dz_a dz_b = m dm d\rho dz du. \quad (128)$$

The full labeled child integral in (127) is twice the integral over the canonical orientation: one labeled assignment has $z_b - z_a \in \mathcal{U}_+$ and the other has its negative. Rewriting the displayed measure on the one-orientation canonical chart therefore cancels the factor $1/2$. On that chart, Lemma 6.5 gives

$$dm_a dm_b dz_a dz_b = m dm d\rho dz du, \quad (129)$$

so (126) equals $1/m$ times the canonically oriented pair measure. This produces (53).

If both u and $-u$ are retained as distinct split marks, the same physical split has two coordinate representations and the corresponding density convention must compensate for that duplication. The final physical rate is unchanged. This is why the paper fixes a canonical half-space at the outset.

C Forward path density and the fixed ELBO terms

Let split events occur at times t_j^S with marks (i_j, ρ_j, u_j) , and let deletion events occur at t_k^D with deleted component i_k . Relative to the forward event times and split marks, the forward log density is

$$\begin{aligned} \log q_X^F(H) = & - \int_0^\tau \mathbf{1}\{N(X_t) < N_{\max}\} N(X_t) \beta(t) dt + \sum_j \left[\log \beta(t_j^S) + \log \kappa_{t_j^S}(\rho_j, u_j \mid X_{t_j^-}, i_j) \right] \\ & - \int_\tau^T N(X_t) \delta(t) dt + \sum_k \log \delta(t_k^D). \end{aligned} \quad (130)$$

The needed change of variables is the joint map $\Psi : (X, H) \mapsto R$ from (57). We now give its continuous determinant. On a forward flow segment with $n \geq 1$ live components, let $c = \sum_i m_i$ at the start and let G be the integrated evaporation rate. The segment maps the total condensed mass to

$$f_G(c) = \frac{Mce^{-G}}{M - c + ce^{-G}}, \quad \alpha_G(c) = \frac{f_G(c)}{c}. \quad (131)$$

Since $m'_i = \alpha_G(c)m_i$, the determinant on the n mass coordinates is

$$J_{\text{flow}}(c, G, n) = \alpha_G(c)^{n-1} f'_G(c), \quad f'_G(c) = \frac{M^2 e^{-G}}{(M - c + ce^{-G})^2}. \quad (132)$$

To see this directly, $n-1$ directions preserve c and are multiplied by α_G ; the remaining radial direction changes c by $f'_G(c)$.

Fix a discrete lineage pattern. Let $\mathcal{I}_F$ be its nonempty flow segments, let $\mathcal{E}_S$ index its split events, and let m_j^{parent} be the parent mass immediately before split j . Composing the flow maps, split maps, deletions, and event-time reversal gives

$$J_\Psi(X, H) = \prod_{\ell \in \mathcal{I}_F} [\alpha_{G_\ell}(c_\ell)^{n_\ell-1} f'_{G_\ell}(c_\ell)] \prod_{j \in \mathcal{E}_S} m_j^{\text{parent}}. \quad (133)$$

An empty flow segment contributes one. Lemma 6.5 gives each parent-mass factor. Deletion merely moves one component’s coordinates into its future birth mark, so its determinant is one. Reordering events and replacing t by $T - t$ also have absolute determinant one. The derivative is block triangular when event times are retained as coordinates, so their effect on later flow states does not alter the product of diagonal blocks. Canonical orientation removes the duplicate discrete representation. On each canonical lineage chart, this determinant is the coordinate form of the Radon–Nikodym factor used in Lemma 7.1.

Equations (130) and (133), together with any reservoir dequantization density, supply all fixed terms in Theorem 7.2. They are not needed for parameter gradients but must be included in a numerical ELBO.

Although $\delta(t)$ diverges, the expression is finite for every realized history because all event times are strictly below T , and after the last deletion $N(X_t) = 0$. Equivalently, one may evaluate the deletion contribution directly using the uniform deletion-time density $1/(T - \tau)$ for each component alive at τ .

D Derivation of the point-process loss

Conditional on L , the ordered Stage-N birth times have joint density $L!/S_N^L$. Birth marks are sampled sequentially from $A_\theta^{(m)}$. In Stage C, the usual marked point-process survival calculation gives the reverse-record log density

$$\begin{aligned} \log \mathcal{L}_\theta(R) = & \log \Pi_\theta(L \mid M, y) + \log(L!) - L \log S_N + \sum_{e \in \mathcal{E}_B} \log A_\theta^{(m)}(m_e, z_e \mid X_{s_e^-}, s_e, L) \\ & - \int_{S_N}^T K_\theta^{\text{tot}}(X_s, s) ds + \sum_{e \in \mathcal{E}_C} \log K_\theta(a_e, b_e \mid X_{s_e^-}, s_e). \end{aligned} \quad (134)$$

The negative of the θ -dependent terms is exactly (69). The fixed birth-time density explains why no learned Stage-N survival integral appears there.

E A finite-state validation model

The exact experiment in Section 12 can use the following finite analogue.

Let total mass be an integer M_{int} , features lie on a lattice $\{1, \dots, K_{\text{lat}}\}$, and component masses are positive integers. Restrict cardinality to N_{max} and use only discrete split marks that keep masses and features on these lattices. Enumerate every unordered multiset

$$X = (v; \{(m_i, z_i)\}_{i=1}^n) \quad (135)$$

with integer entries satisfying the mass constraint. Set $\gamma \equiv 0$ in this validation toy, because the continuous evaporation ODE does not preserve integer masses. Define the *forward* generator G_t using only the discrete split and deletion maps. Births and merges will appear in its time reversal rather than being added to the forward generator.

Use the row-vector convention for the marginal probability vector P_t :

$$\frac{dP_t}{dt} = P_t G_t. \quad (136)$$

This finite ODE can be integrated directly for a time-dependent schedule. On an interval where G_t is constant, its exact update is $P_{t+\Delta} = P_t \exp(\Delta G_t)$. The singular deletion schedule is not piecewise constant, so it should be handled by the ODE or its known integrated hazard, not by one constant-matrix exponential. Compute reverse rates for $t < T$ and take the all-reservoir terminal law as the limit $t \uparrow T$. The exact reverse off-diagonal generator is

$$\overline{G}_{T-t}(Y, X) = G_t(X, Y) \frac{P_t(X)}{P_t(Y)}, \quad X \neq Y, P_t(Y) > 0, \quad (137)$$

and diagonal entries make rows sum to zero. This benchmark can verify flux reversal, pair counting, the training objective, endpoint distributions, and conservation without numerical density estimation. Because it

uses counting measure, it cannot validate the continuous $1/m$ Jacobian. That factor is checked separately by sampling smooth test functions on both sides of the change-of-variables identity in Lemma 6.5 and comparing Monte Carlo integrals.

F Implementation checklist

Before running large experiments, an implementation should pass the following tests.

1. **Map inversion:** randomly split a component and merge its children; recover the original state to numerical tolerance.
2. **Mass:** after every flow and event, verify $|v + \sum_i m_i - M|/M < 10^{-10}$ in double precision.
3. **Permutation:** permute input components and verify invariant total rates and correspondingly permuted pair probabilities.
4. **Terminal state:** simulate many forward paths and verify that all are empty at T .
5. **Point-process likelihood:** on a constant-rate toy process, compare the implemented path likelihood with the analytic exponential waiting-time likelihood.
6. **Integral estimator:** compare Monte Carlo survival integrals with dense numerical quadrature on stored trajectories.
7. **Coordinate Jacobians:** compare automatic or finite-difference determinants with (12), (49), and (132); separately verify the continuous $1/m$ change of variables by Monte Carlo integration.
8. **Birth bridge:** conditional on each tested L , verify that exactly L births occur and that their ordered times match the order statistics of L independent uniform variables.
9. **Finite reverse:** compare learned and analytical reverse generators in Appendix E.
10. **Thinning:** verify every proposed envelope exceeds the evaluated rate. An invalid envelope biases samples.
11. **Genealogy bookkeeping:** reverse every simulated forward history and check that it returns exactly to X_0 before any neural model is introduced.

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